

CO 1 ///// -- 1 Congenital Talipes Equinovarus CTEV / Clubfoot Introduction Definition CTEV is a complex congenital deformity of the foot characterized by four distinct components Equinus plantarflexion at the ankle Varus inversion of the subtalar joint Adduction of the forefoot and Cavus exaggerated medial longitudinal arch It is commonly remembered by the mnemonic CAVE **Clinical Features** The foot is turned downwards and inwards resembling a club Deep creases are visible on the medial border and posterior aspect of the heel The heel is small and empty calcaneus is pulled up The calf muscles especially on the affected side are visibly atrophied and underdeveloped It is a rigid deformity that cannot be passively corrected to a neutral position Can be unilateral or bilateral **Investigations** Clinical Diagnosis Diagnosis is primarily clinical immediately at birth X rays AP and Lateral views of the foot in stress dorsiflexion Used to assess the severity and document the baseline bony alignment Kites angle talocalcaneal angle X rays are also used to monitor the progress of correction **Complications** If untreated the child walks on the dorsolateral aspect of the foot Formation of painful large callosities and bursa over the weight bearing area Awkward limping gait Rocker bottom foot iatrogenic complication if the midfoot is forced into dorsiflexion before the heel equinus is corrected Severe psychological and cosmetic impact **Medical / Conservative Management** Ponseti Method Gold Standard Serial manipulation and casting starting within the first week of life The deformities are corrected in a specific order Cavus Adductus Varus and finally Equinus Casts are changed weekly Splinting After full correction the foot is maintained in a Denis Browne splint or Foot Abduction Brace full time for 3 months then at night until 4 years of age to prevent relapse **Surgical Management** Percutaneous Tendoachilles Tenotomy Often required as the final step of the Ponseti method to correct residual equinus Posteromedial Soft Tissue Release PMSTR Done for resistant or neglected cases in children 6 months to 2 years involving surgical release of tight medial and posterior ligaments/tendons Bony Surgeries For older neglected children e.g wedge resection triple arthrodesis or Ilizarov ring fixator

--- 2 Congenital Dislocation of Hip CDH / DDH Introduction Definition Now correctly termed Developmental Dysplasia of the Hip DDH this represents a spectrum of hip abnormalities ranging from a shallow dysplastic acetabulum to subluxation and completely dislocated femoral head **Clinical Features** Neonates/Infants Asymmetrical thigh and gluteal skin folds Limited abduction of the affected hip Ortolani Test Positive A palpable clunk as the dislocated hip is reduced back into the socket during abduction Barlow Test Positive The hip can be dislocated out of the socket with posterior pressure and adduction Older Children Walking age Painless limp Shortening of the affected limb Galeazzi sign positive Waddling gait if bilateral or Trendelenburg lurch Trendelenburg test is positive pelvis drops on the opposite side when standing on the affected leg **Investigations** Ultrasound Infants 6 months The investigation of choice because the femoral head is not yet ossified Used to assess acetabular depth and hip stability Grafs classification X rays Children 6 months An AP pelvis radiograph is used Pathological findings include a broken Shenton's line an increased acetabular index and the femoral head lying outside the inner lower quadrant formed by Hilgenreiners and Perkins lines **Complications** Avascular Necrosis AVN of the femoral head often iatrogenic from forced abduction Premature osteoarthritis of the hip in young adulthood Permanent limb length discrepancy and limping Severe limp and compensatory scoliosis **Medical / Conservative Management** Birth to 6 months Pavlik harness A dynamic splint that maintains the hips in flexion and abduction keeping the femoral head centered in the acetabulum to allow it to deepen naturally 6 to 18 months Closed reduction under general anesthesia followed by a Hip Spica cast for 3-6 months **Surgical Management** Open Reduction Indicated if closed reduction fails or for children presenting late 18 months Soft tissues preventing reduction are cleared out Pelvic Osteotomies e.g Salter Pemberton To re orient the shallow acetabulum to better cover the femoral head Femoral Derotation/Shortening Osteotomies To relieve pressure on the joint and prevent AVN

--- 3 Torticollis Congenital Muscular Torticollis Introduction Definition Congenital Muscular Torticollis Wry neck is a deformity characterized by a lateral tilt of the head to one side and rotation of the chin to the opposite side caused by unilateral fibrosis and shortening of the sternocleidomastoid SCM muscle **Clinical Features** Head is tilted toward the side of the contracted muscle and the chin points away from it In the first few weeks of life a firm painless swelling sternomastoid tumor or olive may be palpable in the belly of the SCM muscle This mass resolves over months but leaves a fibrous band Restricted neck range of motion Plagiocephaly Flattening of the face and skull on one side due to consistent sleeping posture if left untreated **Investigations** Clinical Diagnosis is primarily clinical Ultrasound Can confirm the fibrotic mass within the SCM muscle X rays of Cervical Spine Mandatory to rule out bony congenital anomalies of the spine like Klippel Feil syndrome or hemivertebrae that can also cause a wry neck **Complications** Permanent facial asymmetry hemi hypoplasia of the face Compensatory cervical scoliosis Vision problems strabismus due to abnormal head posture Psychological issues due to cosmetic deformity **Medical / Conservative Management** Physiotherapy Gentle passive stretching exercises of the neck performed by parents multiple times a day Positioning Placing toys or positioning the crib so the child is forced to look away from the preferred side actively stretching the muscle Over 90% of cases resolve completely with early conservative management within the first year of life **Surgical Management** Indications Failure of conservative treatment after 1 year of age or presentation in an older child Procedure Surgical release tenotomy of the sternocleidomastoid muscle unipolar release at the clavicle/sternum or bipolar release including the mastoid attachment Followed by a corrective brace/collar

and aggressive physiotherapy --- **4 Scoliosis Congenital and Idiopathic Introduction Definition** Scoliosis is defined as a complex 3 dimensional deformity of the spine characterized by a lateral curvature in the coronal plane exceeding 10 degrees accompanied by vertebral rotation Congenital scoliosis is caused by bony anomalies like hemivertebrae while idiopathic is of unknown cause **Clinical Features** Usually painless pain is a red flag for tumors or tethered cord Asymmetrical shoulder height Prominent or protruding scapula on one side Asymmetry of the waistline creases Adams Forward Bending Test Pathognomonic sign is a rib hump on the convex side of the curve caused by vertebral rotation pushing the ribs backward **Investigations** X rays Standing AP and Lateral of whole spine Confirms diagnosis The severity of the curve is measured using Cobbs angle MRI of the Spine Essential especially in congenital or early onset scoliosis to rule out intra spinal anomalies syringomyelia tethered cord Arnold Chiari malformation **Complications** Cardiopulmonary compromise Cor Pulmonale Severe thoracic curves 90 degrees severely restrict lung capacity leading to right heart failure and reduced life expectancy Severe cosmetic deformity and psychological distress Back pain and degenerative arthritis in adulthood Neurological deficits rare but possible in rapidly progressive congenital curves **Medical / Conservative**

Management Observation Curves 20 degrees are observed with serial X rays every 6 months Bracing e.g Boston brace Milwaukee brace Indicated for flexible curves between 20 to 40 degrees in growing children Braces do not correct the curve but aim to halt its progression until skeletal maturity **Surgical Management** Indications Curves 45 50 degrees documented rapid progression or significant cosmetic/respiratory impairment Procedure Posterior Spinal Fusion with instrumentation pedicle screws hooks and rods to correct the curve and fuse the spine in the corrected position In congenital scoliosis hemivertebra excision and short segment fusion may be done early --- **5 Flat Foot Pes Planus Introduction Definition** Flat foot is the loss or collapse of the medial longitudinal arch of the foot resulting in the medial border of the foot coming into contact with the ground It can be flexible arch appears when non weight bearing or on tiptoes or rigid arch is absent constantly **Clinical Features** Medial border of the foot touches the floor when standing Heel is usually in a valgus position outward deviation Flexible Flat Foot Arch reconstitutes when the patient stands on their tiptoes or when the great toe is passively dorsiflexed Jacks test Usually painless Rigid Flat Foot Arch does not reconstitute Often painful with severe restriction of subtalar joint movement Commonly caused by tarsal coalition fusion of foot bones **Investigations** Weight bearing X rays AP and Lateral Lateral view shows the collapse of the arch Mearys angle angle between the talus and first metatarsal is increased CT Scan The best modality to diagnose a tarsal coalition in rigid flat feet **Complications** Foot fatigue and cramping in the calves after walking or running Abnormal wear of shoes wearing out the inner border Predisposition to developing Hallux Valgus bunions or Plantar Fasciitis Chronic foot pain and stiffness in rigid types **Medical / Conservative**

Management Flexible Asymptomatic Reassurance only It is a normal variant in toddlers No treatment is needed Flexible Symptomatic Custom arch supports orthotics medial heel wedges and Achilles tendon stretching exercises to relieve fatigue Rigid Immobilization in a cast or use of rigid orthotics to reduce inflammation and pain **Surgical Management** Indications Severe painful flat feet failing conservative treatment Flexible Calcaneal lengthening osteotomies or tendon transfers repair of tibialis posterior Rigid Tarsal Coalition Resection of the coalition block If severe arthritis is present a triple arthrodesis fusion of the subtalar talonavicular and calcaneocuboid joints is performed --- **6 Vertical Talus Congenital Convex Pes**

Valgus Introduction Definition Congenital Vertical Talus is a rare severe and rigid flatfoot deformity present at birth The talus bone is fixed in an extreme vertical plantarflexed position and the navicular bone is dorsally dislocated over the neck of the talus locking it in place **Clinical Features** Rocker bottom deformity The sole of the foot is convex rather than concave The forefoot is dorsiflexed and abducted The hindfoot heel is fixed in equinus pulled up The head of the talus forms a prominent hard peg like bump on the plantar medial aspect of the sole The deformity is rigidly fixed **Investigations** Lateral X ray in maximum plantarflexion This is diagnostic In a vertical talus the navicular or the first metatarsal axis if the navicular is not ossified does not align with the axis of the talus even when the foot is forced into plantarflexion **Complications** If untreated the child walks directly on the head of the talus Formation of massive painful callosities and ulcerations on the sole Inability to wear normal shoes Severe permanent gait disability and foot pain **Medical / Conservative**

Management Unlike flexible flat feet or clubfeet conservative management alone is rarely curative Serial Casting The reverse Ponseti casting method is used initially to stretch the dorsal and lateral soft tissues to make eventual surgery less extensive **Surgical Management** Surgery is almost always required ideally before the child reaches walking age 6 12 months Open Reduction Extensive soft tissue releases reduction of the talonavicular joint and pinning with Kirschner wires K wires Minimally Invasive Dobbs method serial casting followed by percutaneous pinning and Achilles tenotomy Salvage Talectomy removal of talus or triple arthrodesis in severely neglected older children --- **7 Hand Anomalies Syndactyly Polydactyly Ectrodactyly Introduction Definition** Syndactyly Failure of separation of adjacent digits webbed fingers Polydactyly Presence of extra supernumerary digits Ectrodactyly Cleft hand or lobster claw deformity characterized by a central V shaped cleft and missing central digits **Clinical Features** Syndactyly Digits are fused It can be simple only skin/soft tissue fused or complex bones and joints are fused Can be complete up to the fingertips or incomplete Polydactyly Extra digits can be Preaxial on the thumb/radial side or Postaxial on the little finger/ulnar side Ranges from a floppy skin tag to a fully

functional digit with its own joints and tendons Ectrodactyly Missing middle finger/metacarpals with the remaining fingers separated by a deep central cleft severely altering hand appearance **Investigations** Plain X rays of the Hand Mandatory to define the bony anatomy confirm if joints are shared complex syndactyly and see if the extra digit in polydactyly has bony support or is just a soft tissue appendage **Complications** Severe psychological and social stigma due to cosmetic appearance Reduced hand span and grip strength In syndactyly if digits of unequal length e.g. thumb and index are tethered together it restricts the growth of the longer digit leading to joint deviation and contractures **Medical / Conservative**

Management Occupational therapy for adaptive training and maximizing hand function Splinting to prevent contractures in awaiting surgical candidates **Surgical Management** Syndactyly Surgical separation web space reconstruction using zig zag incisions and full thickness skin grafts Done early around 6-12 months if unequal digits are fused to prevent growth tethering Polydactyly Surgical excision of the extra digit If the extra digit shares a joint with the normal digit careful reconstruction of collateral ligaments and tendons is required to ensure the remaining digit is stable Ectrodactyly Surgical closure of the cleft transposition of digits and soft tissue releases to improve pinch function and cosmetic appearance --- **8**

Arthrogryposis Multiplex Congenita AMC / Amyoplasia Congenita Introduction Definition AMC is a non progressive congenital syndrome characterized by multiple severe rigid joint contractures present at birth in two or more areas of the body It is associated with the replacement of normal skeletal muscle with fibrous and fatty tissue **Clinical Features** Joints Limbs appear wooden or rigid The joints lack normal creases and instead have dimples over them Classic Posture Shoulders internally rotated elbows fixed in extension wrists flexed and ulnarly deviated hips dislocated knees extended and severe rigid clubfeet Sensation and intelligence are completely normal Skin is tense and lacks normal subcutaneous tissue **Investigations** Clinical Strongly clinical diagnosis based on multiple joint involvement X rays To check for joint dislocations especially hips and assess bony alignment Muscle Biopsy/EMG May be done to differentiate from progressive muscular dystrophies or spinal muscular atrophy showing fibrofatty replacement of muscle **Complications** Extreme difficulty with activities of daily living inability to feed oneself if elbows are fixed in extension Inability to walk due to hip dislocations and rigid feet High risk of iatrogenic fractures if joints are manipulated too forcefully during physical therapy Development of secondary scoliosis **Medical / Conservative Management** Must begin immediately after birth Aggressive daily passive stretching and physical therapy to maximize whatever limited joint motion exists Serial casting and custom thermoplastic splints to hold the joints in corrected positions **Surgical Management** Goal Not to restore normal motion which is impossible but to place the limbs in functional positions Upper Limb Soft tissue releases or osteotomies to flex the elbows so the child can reach their mouth to eat and fuse/stabilize the wrists Lower Limb Relocation of hips releasing knee contractures and extensive bony surgeries talectomies osteotomies to make the feet plantigrade so the child can stand --- **9**

Limb Deficiencies Amelia and Phocomelia Introduction Definition These are severe congenital failures of limb formation Amelia The complete absence of one or more limbs transverse deficiency Phocomelia A longitudinal deficiency where the proximal segments of the limb arm/bone are absent or severely hypoplastic but the distal hand or foot is attached directly to the trunk resembling the flippers of a seal Historically linked to the maternal use of Thalidomide **Clinical**

Features Profound visibly obvious absence of limb segments at birth In phocomelia the attached digits may be deformed or functionally impaired but patients often develop extraordinary dexterity with them General health and intellect are usually unaffected though associated organ anomalies can occur **Investigations** X rays To accurately map the residual skeletal anatomy and identify which bones are completely absent versus hypoplastic Systemic Ultrasound/ECHO To rule out associated visceral anomalies cardiac renal gastrointestinal **Complications** Severe impairment of independent mobility and self care Extreme psychological and emotional challenges for the patient and parents Difficulty fitting standard prostheses due to the lack of a normal residual stump Skeletal imbalances leading to severe scoliosis **Medical / Conservative**

Management Intense Occupational/Physical Therapy Training to use remaining limbs or other body parts e.g. using feet for writing or feeding to achieve independence Prosthetics The mainstay of treatment Myoelectric arms or body powered prosthetics Fitting begins early around 6 months so the child integrates the prosthesis into their body image **Surgical**

Management Reconstructive surgery is limited due to the sheer lack of tissue Amputations Occasionally non functional flail digits in phocomelia are amputated to allow a modern prosthesis to fit better over the stump Rotationplasty In specific lower limb deficiencies like PFFD the leg is rotated 180 degrees so the ankle joint functions as a knee joint within a prosthesis --- **10 Klippel Feil Syndrome Introduction Definition** Klippel Feil Syndrome is a congenital malformation characterized by the failure of segmentation fusion of two or more cervical vertebrae **Clinical Features** Classic Triad Present in 50% of cases Short neck Low posterior hairline Markedly restricted range of motion of the neck Webbed Neck Pterygium colli may be visually apparent Sprengels Deformity Congenital undescended high riding scapula is present in about 30% of patients Neurological symptoms are usually absent in childhood but can develop later in life **Investigations** X rays Cervical Spine Shows the block vertebrae fused vertebral bodies absent disc spaces and often associated hemivertebrae MRI Spine Mandatory to check for associated spinal cord anomalies like syringomyelia tethered cord or diastematomyelia

Ultrasound/ECHO To check for common associated anomalies renal agenesis congenital heart

defects **Complications** Cervical Instability The unfused segments adjacent to the fused blocks become hypermobile to compensate This hypermobility leads to early severe degenerative changes and instability Neurological Injury Minor trauma can cause fatal or paralyzing spinal cord compression at the hypermobile segments Deafness common association **Medical / Conservative Management** Activity Restriction Absolute avoidance of contact sports gymnastics or activities with a risk of head/neck trauma Observation and symptomatic treatment NSAIDs gentle physical therapy for mechanical neck pain **Surgical Management** Prophylactic Spinal Fusion Indicated if there is documented dangerous instability at the cervical segments adjacent to the fusion block or if there is progressive neurological deficit Woodward or Green Procedure Surgical relocation of the scapula downwards if an associated Sprengels deformity is causing severe cosmetic or functional impairment --- **11 Osteogenesis Imperfecta Fragile Ossium Introduction Definition** Osteogenesis Imperfecta Brittle Bone Disease is a genetic disorder of connective tissue caused by mutations in the genes COL1A1/COL1A2 responsible for producing Type I collagen This results in severe osteopenia and extreme bone fragility **Clinical Features** Multiple Fractures Occur from minimal or no trauma Even routine diaper changing can cause femur fractures in severe types Blue Sclerae The sclera of the eyes appear blue/gray because the thin scleral collagen allows the underlying choroid pigment to show through Dentinogenesis Imperfecta Teeth are discolored yellow/brown translucent and prone to rapid wear and breakage Deformities Severe bowing of long bones due to microfractures and soft bone Joint hypermobility and hearing loss otosclerosis in adulthood **Investigations** X rays Reveal severe generalized osteopenia extremely thin cortices multiple healing fractures at various stages severe bowing of long bones and sometimes popcorn calcifications in the metaphyses DEXA Scan To quantify the severe reduction in bone mineral density Genetic Testing Confirms the specific collagen mutation **Complications** Severe crippling skeletal deformities leading to wheelchair dependence Extreme short stature dwarfism Non unions or severe malunions of fractures Basilar Invagination The soft skull base settles onto the cervical spine causing life threatening compression of the brainstem and spinal cord **Medical / Conservative Management** Bisphosphonates e g IV Pamidronate or Zoledronic acid are the medical mainstay They inhibit osteoclast activity significantly increasing bone density reducing pain and decreasing the fracture rate Careful handling of the infant lifting by the trunk never by the limbs Lightweight bracing to support bowed limbs **Surgical Management** Intramedullary Rodding Standard surgery for repeated fractures and bowing The bone is cut into segments osteotomies straightened like beads on a string and fixed with an intramedullary rod Telescopic Rods Fassier Duval rods Specialized rods that expand as the child grows preventing the bone from growing past the rod and fracturing at the tip Avoid Plates Traditional plates and screws are avoided as the screws easily pull out of the fragile bone --- **12 Cervical Rib Introduction Definition** A cervical rib is a congenital supernumerary extra rib that arises from the transverse process of the seventh cervical vertebra C7 It can be a fully formed bony rib or a fibrous band attaching to the first rib potentially causing compression in the thoracic outlet **Clinical Features** The vast majority are entirely asymptomatic and found incidentally Thoracic Outlet Syndrome TOS If symptomatic it causes compression of the brachial plexus C8 T1 nerve roots and the subclavian artery/vein Neurological Pain tingling and numbness along the ulnar border of the forearm and hand Weakness clumsiness and wasting of the intrinsic muscles of the hand thenar and hypothenar eminences Vascular The hand becomes cold pale and pulseless especially when the arm is raised Adsons Test Positive radial pulse disappears upon taking a deep breath and turning the head toward the affected side **Investigations** X rays AP Neck/Chest Identifies the presence of a bony cervical rib or an elongated C7 transverse process Nerve Conduction Studies EMG/NCV Confirms neurological compression at the thoracic outlet differentiating it from ulnar neuropathy at the elbow Doppler Ultrasound / Angiography Used if vascular compression or a subclavian artery aneurysm is suspected **Complications** Severe intractable neurogenic pain Permanent wasting and loss of function of hand muscles Vascular Post stenotic dilatation of the subclavian artery leading to aneurysm formation thrombosis and distal embolization causing gangrene of the fingertips **Medical / Conservative Management** First line treatment for purely neurological mild symptoms Physiotherapy Specific exercises aimed at strengthening the shoulder girdle muscles trapezius levator scapulae to elevate the shoulder and open up the thoracic outlet space Postural correction avoiding slouching NSAIDs for pain relief **Surgical Management** Indications Failure of 3 6 months of conservative therapy progressive muscle wasting or any signs of vascular compression ischemia aneurysm Procedure Surgical excision of the cervical rib or the fibrous band along with the division of the scalenus anterior muscle scaleneotomy This thoroughly decompresses the brachial plexus and the subclavian artery /////////////// --- **1 ACQUIRED TORTICOLLIS DEFINITION** Acquired torticollis is a condition in which the head assumes an abnormal position due to shortening spasm pain or dysfunction of the cervical muscles producing lateral flexion of the head to one side with rotation of the chin toward the opposite side **CAUSES** • Muscular causes • Sternocleidomastoid spasm • Cervical muscle strain • Myofascial pain • Trauma • Whiplash injury • Cervical fracture/dislocation • Infection • Retropharyngeal abscess • Cervical tuberculosis • Neurological causes • Dystonia • Cerebral lesions • Ocular causes • Abnormal ocular alignment • Cervical spine disorders • Cervical spondylosis • Disc prolapse • Inflammatory conditions • Rheumatoid arthritis • Postural causes **CLINICAL FEATURES** • Head tilted to one side • Chin rotated toward opposite side • Neck pain • Cervical muscle spasm • Restricted cervical ROM •

Tenderness • Shoulder asymmetry • Difficulty turning the head • Headache may occur • In chronic cases facial asymmetry may develop **COMPLICATIONS** • Persistent neck pain • Cervical stiffness • Muscle shortening • Postural deformity • Headache • Secondary scoliosis • Functional limitation **INVESTIGATIONS** • Cervical spine X ray • CT/MRI when structural pathology is suspected • Neurological examination • Blood investigations when infection/inflammation is suspected **CONSERVATIVE/MEDICAL MANAGEMENT** • Treatment of underlying cause • Analgesics/NSAIDs • Heat therapy • Gentle cervical ROM • Stretching of shortened muscles • Strengthening of weak muscles • Postural correction • Cervical traction in selected cases • Soft cervical collar for short term use when appropriate • Botulinum toxin for selected dystonic torticollis **SURGICAL MANAGEMENT** Indicated when severe fixed deformity or structural pathology persists • Sternocleidomastoid release/lengthening • Muscle/tendon release • Cervical decompression or stabilization when indicated by underlying pathology **FLOW CHART** Underlying cause muscle spasm/shortening abnormal head position restricted ROM chronic deformity functional limitation --- **2 ACQUIRED SCOLIOSIS DEFINITION** Scoliosis is a three dimensional deformity of the spine characterized by lateral curvature associated with vertebral rotation generally measured by the Cobb angle on radiographs Acquired scoliosis develops after birth due to a known or secondary cause **CAUSES** Poor posture Leg length discrepancy Muscle imbalance Cerebral palsy Muscular dystrophy Poliomyelitis Spinal tuberculosis Vertebral fracture Tumors Degenerative spinal disease Painful conditions Trauma **CLINICAL FEATURES** Unequal shoulder height Unequal scapular prominence Unequal waist contours Pelvic asymmetry Lateral spinal curvature Rib prominence Trunk shift Back pain particularly in degenerative scoliosis Reduced spinal mobility Leg length discrepancy may be present **CLINICAL TEST** Adams forward bending test Patient bends forward Rib hump or lumbar prominence suggests vertebral rotation **COMPLICATIONS** Chronic back pain Progressive deformity Reduced pulmonary function in severe curves Reduced exercise tolerance Neurological compromise in selected structural causes Psychological/social problems Functional disability **RADIOLOGICAL FEATURES** Standing AP/PA and lateral spine radiographs demonstrate Lateral curvature Vertebral rotation Cobb angle Vertebral wedging Degenerative changes where applicable **CONSERVATIVE MANAGEMENT** Treat underlying cause • Physiotherapy • Postural correction • Core strengthening • Spinal mobility exercises • Stretching • Breathing exercises • Bracing in selected growing patients • Leg length correction if discrepancy contributes • Analgesics for pain **SURGICAL MANAGEMENT** Indications depend on severity progression symptoms and cause • Spinal instrumentation and fusion • Corrective osteotomy in selected rigid deformities • Decompression if neurological compression exists • Treatment of underlying tumor/infection when responsible **FLOW CHART** Underlying condition muscle imbalance/vertebral pathology lateral curvature rotation progressive deformity pain/functional impairment --- **3 KYPHOSIS DEFINITION** Kyphosis is an excessive posterior convex curvature of the thoracic or thoracolumbar spine producing an exaggerated rounded back appearance **CAUSES** • Poor posture • Osteoporosis • Vertebral compression fractures • Scheuermann disease • Tuberculosis of spine • Degenerative disease • Trauma • Tumor • Neuromuscular disorders • Ankylosing spondylitis **CLINICAL FEATURES** • Rounded back • Forward head posture • Increased thoracic curvature • Shoulder protraction • Back pain • Hamstring tightness may occur • Reduced spinal mobility • Reduced chest expansion in severe cases • Fatigue • Neurological symptoms in severe structural disease **COMPLICATIONS** • Chronic pain • Progressive deformity • Restrictive respiratory impairment • Reduced balance • Increased fall risk • Neurological compression • Spinal cord injury in severe cases **RADIOLOGICAL FEATURES** • Increased thoracic kyphotic angle • Vertebral wedging • Compression fractures • Degenerative changes • Vertebral destruction in infective disease **CONSERVATIVE MANAGEMENT** • Postural education • Thoracic extension exercises • Back extensor strengthening • Abdominal strengthening • Stretching of pectoral muscles • Breathing exercises • Analgesics • Osteoporosis management • Bracing in selected patients **SURGICAL MANAGEMENT** • Vertebral augmentation in selected painful compression fractures • Corrective osteotomy • Spinal fusion • Decompression and stabilization when neurological compression/instability is present --- **4 LORDOSIS DEFINITION** Lordosis is an excessive anterior convex curvature of the spine most commonly affecting the lumbar region **CAUSES** • Obesity • Pregnancy • Weak abdominal muscles • Tight hip flexors • Spondylolisthesis • Hip flexion deformity • Osteoporosis • Spinal deformity • Poor posture • Neuromuscular disorders **CLINICAL FEATURES** • Increased lumbar curvature • Anterior pelvic tilt • Protruding abdomen • Prominent buttocks • Low back pain • Tight hip flexors • Weak abdominal/gluteal muscles • Hamstring dysfunction may occur **COMPLICATIONS** • Chronic low back pain • Facet joint stress • Muscle fatigue • Reduced mobility • Secondary disc/facet degeneration • Neurological symptoms if associated with spinal pathology **CONSERVATIVE MANAGEMENT** • Postural correction • Abdominal strengthening • Gluteal strengthening • Hip flexor stretching • Hamstring stretching when appropriate • Core stabilization • Weight management • Ergonomic modification • Analgesia **SURGICAL MANAGEMENT** Surgery is directed at the underlying cause • Decompression • Spinal stabilization/fusion • Corrective osteotomy • Treatment of spondylolisthesis when indicated --- **5 GENU VARUM DEFINITION** Genu varum or bow legs is a deformity in which the knees remain separated when the ankles are brought together producing an outward angulation of the legs **CAUSES** • Physiological deformity • Rickets • Blount disease • Malunited fractures • Growth plate injury •

Osteoarthritis • Skeletal dysplasia • Obesity in some cases **CLINICAL FEATURES** • Bowing of legs • Knees separated when ankles touch • Altered gait • Lateral thrust at knee • Knee pain in severe cases • Unequal loading of knee joint **COMPLICATIONS** • Medial compartment knee overload • Early osteoarthritis • Patellofemoral dysfunction • Gait abnormalities • Progressive deformity **RADIOLOGY** • Increased medial angulation • Assessment of mechanical axis • Growth plate abnormalities where applicable • Degenerative changes in adults **CONSERVATIVE MANAGEMENT** • Treat underlying metabolic disease • Weight management • Physiotherapy • Muscle strengthening • Gait training • Orthoses in selected cases **SURGICAL MANAGEMENT** For significant persistent/progressive deformity • Guided growth/hemiepiphyodesis in growing children • Corrective osteotomy in mature patients • Knee replacement for severe degenerative disease --- **6 GENU VALGUM DEFINITION** Genu valgum or knock knee deformity is a condition in which the knees approximate or touch while the ankles remain separated **CAUSES** • Physiological childhood valgus • Rickets • Obesity • Growth plate injury • Malunion • Skeletal dysplasia • Osteoarthritis • Previous trauma **CLINICAL FEATURES** • Knees come together • Ankles remain apart • Altered gait • Patellar maltracking • Knee pain • Difficulty running/walking in severe cases **COMPLICATIONS** • Patellofemoral instability • Lateral compartment overload • Early osteoarthritis • Gait abnormalities • Functional limitation **CONSERVATIVE MANAGEMENT** • Treatment of underlying cause • Weight management • Hip and knee strengthening • Gait training • Orthoses in selected cases **SURGICAL MANAGEMENT** • Guided growth • Distal femoral osteotomy • Corrective osteotomy • Joint replacement in severe degenerative disease --- **7 GENU RECURVATUM DEFINITION** Genu recurvatum is a deformity in which the knee is excessively extended beyond the normal physiological range **CAUSES** • Ligamentous laxity • Neuromuscular disorders • Poliomyelitis • Quadriceps weakness • Anterior capsular laxity • Trauma • Malunion • Abnormal tibial slope • Poor gait mechanics **CLINICAL FEATURES** • Hyperextension of knee • Posterior displacement of knee during standing • Knee pain • Instability • Abnormal gait • Quadriceps weakness • Difficulty controlling knee during stance **COMPLICATIONS** • Chronic ligament strain • Knee instability • Pain • Degenerative changes • Meniscal injury • Gait dysfunction **CONSERVATIVE MANAGEMENT** • Quadriceps and hamstring strengthening • Proprioceptive training • Gait retraining • Knee brace/orthosis • Correction of underlying neuromuscular disorder • Activity modification **SURGICAL MANAGEMENT** • Ligament reconstruction where indicated • Corrective osteotomy • Growth modulation in selected children • Other reconstruction based on structural cause --- **8 COXA VARA DEFINITION** Coxa vara is a deformity in which the femoral neck shaft angle is reduced generally below approximately 120 in adults **CAUSES** • Congenital/developmental conditions • Rickets • Trauma • Fracture malunion • Fibrous dysplasia • Perthes disease • Metabolic bone disease **CLINICAL FEATURES** • Shortening of limb • Limp • Trendelenburg gait • Reduced hip abduction • Apparent shortening • Hip pain • Muscle weakness **COMPLICATIONS** • Progressive deformity • Hip instability • Early osteoarthritis • Limb length discrepancy • Abductor weakness **RADIOLOGY** • Reduced neck shaft angle • Shortened/vertical femoral neck • Altered femoral head mechanics **CONSERVATIVE MANAGEMENT** • Treat underlying metabolic disorder • Physiotherapy • Hip abductor strengthening • Gait training • Activity modification **SURGICAL MANAGEMENT** • Corrective valgus osteotomy • Internal fixation • Treatment of underlying bone pathology --- **9 COXA VALGA DEFINITION** Coxa valga is a deformity in which the femoral neck shaft angle is increased generally above approximately 135 in adults **CAUSES** • Neuromuscular disorders • Developmental abnormalities • Hip dysplasia • Cerebral palsy • Muscle imbalance • Certain metabolic/skeletal disorders **CLINICAL FEATURES** • Reduced hip stability • Limp • Abductor weakness • Limited hip mechanics • Possible hip pain • Increased tendency toward subluxation in susceptible patients **COMPLICATIONS** • Hip instability • Subluxation/dislocation • Abductor insufficiency • Altered gait • Early degenerative changes **MANAGEMENT** • Treat underlying condition • Hip strengthening • Gait training • Orthotic management when indicated • Corrective femoral osteotomy in significant symptomatic deformity • Hip reconstruction when associated with instability/dysplasia --- **10 PES CAVUS DEFINITION** Pes cavus is a foot deformity characterized by an abnormally high medial longitudinal arch often associated with plantarflexion of the first ray hindfoot varus and clawing of the toes **CAUSES** • Neuromuscular disorders • Hereditary motor sensory neuropathy • Poliomyelitis • Cerebral palsy • Spinal cord disorders • Trauma • Idiopathic causes **CLINICAL FEATURES** • High medial arch • Shortened foot appearance • Hindfoot varus • Claw toes • Prominent metatarsal heads • Lateral border weight bearing • Callosities • Foot pain • Ankle instability **COMPLICATIONS** • Metatarsalgia • Calluses • Ankle sprains • Plantar fascia pain • Difficulty wearing shoes • Progressive deformity • Gait abnormality **CONSERVATIVE MANAGEMENT** • Treat underlying neurological disease • Custom orthoses • Appropriate footwear • Stretching • Strengthening • Balance training • Gait correction • Pressure redistribution **SURGICAL MANAGEMENT** • Plantar fascia release in selected flexible deformities • Tendon transfer • Calcaneal osteotomy • First ray osteotomy • Midfoot osteotomy • Arthrodesis for rigid severe deformity --- **11 PES PLANUS DEFINITION** Pes planus or flat foot is a condition in which the medial longitudinal arch of the foot is reduced or absent often accompanied by excessive hindfoot valgus and pronation **CAUSES** • Ligamentous laxity • Posterior tibial tendon dysfunction • Obesity • Trauma • Rheumatoid arthritis • Neuromuscular disorders • Degenerative changes • Tarsal coalition **CLINICAL FEATURES** • Flattened medial arch • Excessive pronation • Hindfoot valgus • Heel eversion • Forefoot

abduction • Foot fatigue • Medial foot/ankle pain • Difficulty prolonged standing/walking • Abnormal shoe wear

COMPLICATIONS • Posterior tibial tendon dysfunction • Plantar fasciitis • Ankle pain • Knee/hip biomechanical problems • Progressive deformity • Difficulty walking

CONSERVATIVE MANAGEMENT • Weight management • Footwear modification • Arch support orthoses • Heel support • Stretching of tight calf muscles • Foot intrinsic strengthening • Posterior tibial muscle strengthening • Balance and gait training • Activity modification

SURGICAL MANAGEMENT For symptomatic rigid/progressive deformity • Tendon reconstruction • Calcaneal osteotomy • Midfoot reconstruction • Arthrodesis for rigid deformity • Procedures addressing the underlying cause ---

12 HALLUX RIGIDUS DEFINITION Hallux rigidus is a degenerative arthritic condition of the first metatarsophalangeal joint characterized predominantly by painful restriction of great toe dorsiflexion

CAUSES • Degenerative osteoarthritis • Previous trauma • Repetitive loading • Anatomical abnormalities • Inflammatory arthritis • Previous surgery

CLINICAL FEATURES • Pain at first MTP joint • Restricted dorsiflexion • Difficulty during push off • Dorsal osteophyte • Swelling • Stiffness • Altered gait • Shoe discomfort

COMPLICATIONS • Progressive joint degeneration • Gait alteration • Transfer metatarsalgia • Callus formation • Reduced walking tolerance

RADIOLOGY • Joint space narrowing • Dorsal osteophytes • Subchondral sclerosis • Cystic changes • Advanced joint destruction

CONSERVATIVE MANAGEMENT • Activity modification • NSAIDs/analgesics • Stiff soled shoes • Rocker bottom footwear • Orthoses • Physiotherapy • ROM exercises within tolerance • Intra articular injection in selected cases

SURGICAL MANAGEMENT • Cheilectomy for selected early disease • Osteotomy in selected patients • First MTP arthrodesis for advanced disease • Joint replacement in selected patients ---

13 HALLUX VALGUS DEFINITION Hallux valgus is a progressive deformity in which the great toe deviates laterally toward the lesser toes with medial deviation of the first metatarsal producing a prominent medial eminence or bunion

CAUSES/RISK FACTORS • Genetic predisposition • Footwear with narrow toe box • Ligamentous laxity • Pes planus • Abnormal first ray mechanics • Rheumatoid arthritis • Neuromuscular disorders

CLINICAL FEATURES • Lateral deviation of great toe • Medial bunion • Pain • Redness and pressure over bunion • Difficulty wearing shoes • Callosity • Reduced first MTP movement • Transfer metatarsalgia • Lesser toe deformity

COMPLICATIONS • Bursitis • Osteoarthritis • Metatarsalgia • Hammer/claw toes • Skin ulceration • Persistent gait disturbance

RADIOLOGICAL FEATURES Weight bearing X ray may show • Increased hallux valgus angle • Increased intermetatarsal angle • Medial deviation of first metatarsal • Subluxation of first MTP joint • Degenerative changes

CONSERVATIVE MANAGEMENT • Wide toe box footwear • Avoid tight/high heeled shoes • Bunion pads • Orthoses • Toe spacers in selected cases • Analgesics • Physiotherapy • Calf/foot stretching and strengthening • Activity modification

SURGICAL MANAGEMENT Indicated for persistent pain and functional limitation despite conservative treatment • Distal metatarsal osteotomy • Proximal metatarsal osteotomy • Lapidus procedure in selected patients • Akin osteotomy • First MTP arthrodesis for severe arthritis/deformity • Bunion correction according to deformity severity ---

14 HAMMER TOE DEFINITION Hammer toe is a deformity of a lesser toe characterized by flexion deformity of the proximal interphalangeal PIP joint often with relative extension at the MTP and DIP joints

CAUSES • Tight footwear • Muscle imbalance • Hallux valgus • Pes cavus • Rheumatoid arthritis • Neurological disorders • Tendon imbalance • Trauma

CLINICAL FEATURES • Flexed PIP joint • Pain over prominent PIP joint • Corn formation • Callosity • Difficulty wearing shoes • MTP pain • Reduced toe mobility • Gait difficulty in severe cases

COMPLICATIONS • Corns • Ulceration • Infection • MTP instability • Metatarsalgia • Fixed deformity • Difficulty walking

CONSERVATIVE MANAGEMENT • Wide toe box shoes • Avoid tight footwear • Toe pads • Orthoses • Splinting/taping in flexible deformity • Toe stretching • Intrinsic muscle strengthening • Callus care

SURGICAL MANAGEMENT For painful fixed deformity • Tendon release/transfer • PIP joint arthroplasty • PIP arthrodesis • MTP procedures when required • Correction of associated hallux valgus/pes cavus ---

15 METATARSALGIA DEFINITION Metatarsalgia is a clinical condition characterized by pain and tenderness beneath the metatarsal heads usually caused by excessive or abnormal loading of the forefoot

CAUSES • Excessive activity/running • Improper footwear • High heels • Pes cavus • Pes planus • Hallux valgus • Claw/hammer toes • Short or long metatarsals • Fat pad atrophy • Stress fracture • Inflammatory arthritis

CLINICAL FEATURES • Pain beneath metatarsal heads • Burning or aching sensation • Pain worsens during walking/running • Forefoot tenderness • Callosity • Feeling of walking on a pebble • Reduced walking tolerance

COMPLICATIONS • Chronic pain • Altered gait • Callus formation • Lesser toe deformity • Reduced physical activity • Secondary knee/hip biomechanical problems

INVESTIGATIONS • Clinical examination • Weight bearing X ray • Ultrasound/MRI when neuroma stress fracture or soft tissue pathology is suspected

CONSERVATIVE MANAGEMENT • Activity modification • Rest during painful period • Proper footwear with wide toe box • Metatarsal pads • Custom orthoses • Shock absorbing insoles • Calf and foot stretching • Foot intrinsic strengthening • Weight management • Analgesics/NSAIDs when appropriate

SURGICAL MANAGEMENT Surgery is considered when persistent symptoms fail conservative treatment or when a structural cause is identified • Metatarsal osteotomy • Correction of associated hallux valgus • Correction of lesser toe deformity • Excision of neuroma where appropriate/////

CO 2,3 //// 1 OSTEOMYELITIS ACUTE AND CHRONIC **Definition** Osteomyelitis is an inflammatory and infective disease of bone and bone marrow usually caused by microorganisms most commonly bacteria resulting in destruction of bone and surrounding tissues **Acute osteomyelitis** Infection of bone of relatively recent onset usually associated with acute inflammation and systemic manifestations **Chronic osteomyelitis** Persistent or recurrent bone infection lasting for a prolonged period usually associated with necrotic bone sequestrum new bone formation involucrum sinus formation and recurrent discharge **Common Causative Organism** Staphylococcus aureus is the most common organism Other organisms • Streptococci • Gram negative bacilli • Pseudomonas aeruginosa • Salmonella particularly in sickle cell disease • Mycobacterium tuberculosis tuberculous osteomyelitis **Etiology / Causes** 1 Hematogenous spread Microorganisms reach bone through the bloodstream Common in • Children • Metaphysis of long bones • Femur • Tibia • Humerus 2 Direct implantation Organisms are introduced directly into bone following • Open fractures • Penetrating injuries • Orthopaedic surgery • Internal fixation • Joint replacement 3 Contiguous spread Infection spreads from adjacent soft tissue Examples • Diabetic foot ulcer • Pressure sore • Cellulitis • Chronic wound **Predisposing factors** • Diabetes mellitus • Immunosuppression • Malnutrition • Peripheral vascular disease • Trauma • Open fracture • Previous surgery • Prosthetic implants • Intravenous drug use • Chronic skin infection **Pathophysiology** Infection → bacterial multiplication → inflammatory response → vascular congestion → edema → increased intraosseous pressure → vascular compromise → bone necrosis → abscess formation → periosteal elevation → sequestrum formation → involucrum formation → sinus formation Important terms Sequestrum Dead separated piece of infected bone Involucrum New bone formed around a sequestrum by the periosteum Cloaca Opening in the involucrum through which pus or sequestrum may discharge **Clinical Features** **Acute osteomyelitis** Local • Severe localized bone pain • Tenderness • Swelling • Warmth • Redness • Restricted movement of nearby joint • Muscle spasm Systemic • Fever • Chills • Malaise • Irritability • Tachycardia • Loss of appetite **Chronic osteomyelitis** • Persistent bone pain • Recurrent swelling • Discharging sinus • Purulent discharge • Recurrent fever • Non healing wound • Limb deformity • Reduced function • Local tenderness • Skin pigmentation/scarring • Sequestrum may occasionally be palpable **Investigations** **Blood investigations** • CBC leukocytosis in acute infection • ESR raised • CRP raised • Blood culture identifies causative organism • Blood glucose **Imaging** • X ray • Ultrasound for soft tissue collection • MRI highly sensitive for early osteomyelitis • CT useful for cortical destruction and sequestrum • Bone scan where indicated **Microbiological investigation** • Pus/wound culture • Bone biopsy and culture important in chronic or uncertain cases **Complications** **Acute** • Septicemia • Septic shock • Subperiosteal abscess • Soft tissue abscess • Pathological fracture • Growth disturbance in children • Joint infection **Chronic** • Recurrent sinus • Sequestrum • Involucrum • Pathological fracture • Bone deformity • Limb shortening • Secondary amyloidosis • Malignant transformation of chronic sinus classically squamous cell carcinoma Marjolin ulcer • Chronic pain • Functional disability **Medical Management** **Acute osteomyelitis** • Hospitalization when clinically indicated • Blood and tissue cultures • Appropriate intravenous antibiotics initially • Antibiotics modified according to culture and sensitivity • Analgesics • Antipyretics • Adequate hydration and nutrition • Immobilization of affected limb if necessary • Treatment of underlying disease **Chronic osteomyelitis** • Long term culture directed antibiotics • Wound care • Nutritional support • Diabetes control • Treatment of vascular insufficiency • Analgesia • Physiotherapy after acute infection is controlled **Surgical Management** Indications • Abscess • Sequestrum • Failure of antibiotics • Chronic sinus • Necrotic tissue • Infected implant • Pathological fracture Procedures • Incision and drainage • Surgical debridement • Sequestrectomy • Curettage • Saucerization • Removal of infected implants when appropriate • Dead space management • Bone grafting • Local antibiotic delivery • Soft tissue flap coverage • Stabilization of pathological fracture when required **Flow Chart** Infection of bone ↓ Inflammatory response ↓ Edema + increased intraosseous pressure ↓ Vascular compromise ↓ Bone necrosis ↓ Abscess / sequestrum ↓ Chronic infection ↓ Involucrum + sinus formation ↓ Debridement + antibiotics ± reconstruction **Conclusion** Osteomyelitis is a potentially destructive bone infection Early diagnosis appropriate culture directed antibiotics and timely drainage/debridement are essential to prevent chronic infection deformity and functional disability --- **2 BRODIES ABSCESS** **Definition** Brodies abscess is a localized form of subacute osteomyelitis characterized by a localized intraosseous abscess usually occurring in the metaphysis of a long bone especially the tibia The common causative organism is Staphylococcus aureus **Etiology** Usually caused by • Staphylococcus aureus Predisposing factors • Previous minor trauma • Hematogenous infection • Reduced immunity • Diabetes • Previous inadequately treated infection **Common Sites** • Upper end of tibia common • Lower end of femur • Other long bones • Occasionally small bones **Pathophysiology** Bacterial infection ↓ Localized inflammatory response ↓ Intraosseous abscess formation ↓ Granulation tissue around abscess ↓ Reactive sclerosis of surrounding bone Because the infection is localized and relatively slow systemic manifestations may be minimal **Clinical Features** • Localized deep bone pain • Pain may be worse at night • Local tenderness • Mild swelling • Symptoms may persist for weeks or months • Fever usually absent or mild • Restriction of nearby joint movement may occur • Symptoms may be relieved temporarily by salicylates/analgesics **Investigations** **Blood** • CBC may be normal • ESR may be mildly elevated • CRP may be elevated **X**

ray Typical appearance • Well defined radiolucent cavity • Surrounding sclerosis • Usually metaphyseal location **MRI** Shows • Abscess cavity • Bone marrow edema • Soft tissue involvement **CT** Useful for • Defining cortical involvement • Surgical planning **Biopsy** May be required to • Confirm diagnosis • Exclude tumor • Obtain culture **Differential Diagnosis** • Osteosarcoma • Ewing sarcoma • Osteoid osteoma • Chronic osteomyelitis • Bone cyst • Tuberculosis of bone **Complications** • Chronic osteomyelitis • Recurrence • Extension into soft tissues • Pathological fracture uncommon • Growth disturbance in children • Deformity **Medical Management** • Culture directed antibiotics • Analgesics • Rest/temporary immobilization if required • Treatment of predisposing conditions • Follow up imaging **Surgical Management** Surgery may be required when • Abscess is large • Symptoms persist despite antibiotics • Diagnosis is uncertain • There is extensive bone destruction • Recurrence occurs **Procedures** • Curettage of abscess cavity • Drainage • Removal of infected tissue • Cavity filling with bone graft or suitable substitute when indicated • Antibiotic therapy **Flow Chart** Subacute bacterial infection ↓ Localized bone infection ↓ Intraosseous abscess ↓ Surrounding sclerosis ↓ Persistent localized pain ↓ X ray/MRI diagnosis ↓ Antibiotics ± curettage/drainage **Conclusion** Brodie's abscess should be suspected in patients with persistent localized bone pain and a well defined lytic lesion surrounded by sclerosis Early diagnosis prevents progression to chronic osteomyelitis --- **3 TUBERCULOUS ARTHRITIS / TB OF SPINE AND MAJOR**

JOINTS Definition Tuberculosis of bone and joints is a chronic granulomatous infection caused by Mycobacterium tuberculosis usually resulting from hematogenous spread from a primary focus commonly the lungs Tuberculosis may affect • Spine Potts disease • Hip • Knee • Ankle • Shoulder • Elbow • Wrist and other joints **A TUBERCULOSIS OF SPINE POTTS SPINE Definition** Potts spine is tuberculosis of the vertebral column characterized by destruction of vertebral bodies and intervertebral discs often leading to vertebral collapse kyphotic deformity and neurological complications **Etiology** Causative organism • Mycobacterium tuberculosis Usually spreads to the spine through • Hematogenous spread • Lymphatic spread • Extension from adjacent structures **Common Sites** • Lower thoracic spine • Thoracolumbar junction • Lumbar spine **Pathophysiology** TB infection ↓ Granuloma formation in vertebral body ↓ Caseous necrosis ↓ Vertebral destruction ↓ Disc involvement ↓ Vertebral collapse ↓ Kyphosis / gibbus deformity ↓ Cold abscess ± spinal cord compression ↓ Neurological deficit **Clinical Features General** • Low grade fever • Night sweats • Loss of appetite • Weight loss • Malaise **Spinal** • Persistent back pain • Local tenderness • Stiffness • Muscle spasm • Restricted spinal movement • Deformity • Kyphosis • Gibbus deformity **Neurological** • Weakness • Sensory disturbance • Radicular pain • Paraplegia • Bowel/bladder dysfunction in severe disease **Cold abscess** May appear as • Paravertebral swelling • Psoas abscess • Groin swelling **Investigations Blood** • CBC • ESR • CRP • Tuberculosis related microbiological testing as appropriate **Imaging** • X ray spine • MRI spine best for neural compression abscess and soft tissue involvement • CT bony destruction and surgical planning **Microbiological/histological** • Biopsy • Tissue culture • Molecular testing where appropriate **Complications** • Vertebral collapse • Kyphosis • Gibbus deformity • Spinal cord compression • Paraplegia • Nerve root compression • Cold abscess • Sinus formation • Instability • Chronic pain • Severe neurological disability **Medical Management Anti tubercular therapy** Multidrug anti tubercular therapy ATT is the cornerstone of treatment using an appropriate regimen according to current national/institutional guidelines **Other measures** • Analgesics • Nutritional support • Rest during active painful stage • Bracing when appropriate • Management of neurological deficits • Monitoring treatment response **Surgical Management** Indications • Progressive neurological deficit • Significant spinal cord compression • Severe instability • Severe or progressive deformity • Large abscess requiring drainage • Failure of appropriate medical treatment • Mechanical instability **Procedures** • Abscess drainage • Debridement • Decompression • Stabilization • Instrumented fusion • Corrective surgery for severe deformity **B TUBERCULOUS ARTHRITIS OF MAJOR**

JOINTS Definition Tuberculous arthritis is a chronic granulomatous infection of a synovial joint caused by Mycobacterium tuberculosis producing synovial inflammation cartilage destruction and progressive joint damage **Common Joints** • Hip • Knee • Ankle • Shoulder • Elbow • Wrist **Clinical Features** The onset is generally insidious **General** • Low grade fever • Weight loss • Malaise • Reduced appetite **Local** • Pain • Swelling • Stiffness • Reduced range of movement • Muscle wasting • Joint tenderness **Characteristic** • A cold swelling may occur with comparatively little redness or warmth **Advanced disease** • Sinus formation • Joint destruction • Deformity • Subluxation/dislocation • Ankylosis • Limb shortening especially with hip disease **Investigations** • CBC • ESR/CRP • X ray • MRI • Ultrasound for effusion • Joint aspiration • Synovial fluid analysis • AFB testing/culture where appropriate • Molecular tests • Synovial biopsy **Classical radiological findings** The classical description of tuberculous arthritis includes • Periarticular osteoporosis • Peripheral osseous erosions • Gradual joint space narrowing **Complications** • Cartilage destruction • Joint deformity • Subluxation • Dislocation • Ankylosis • Sinus formation • Abscess • Pathological fracture • Limb shortening • Severe functional disability **Medical Management** • Appropriate multidrug ATT • Analgesics • Rest during acute painful stage • Nutritional support • Treatment of associated TB • Physiotherapy after acute inflammation is controlled **Physiotherapy principles** • Maintain safe joint mobility • Prevent contractures • Maintain muscle strength • Gradual functional training • Avoid excessive loading of a severely destroyed joint **Surgical Management** Depending on disease stage • Aspiration/drainage of

abscess • Synovectomy in selected cases • Debridement • Arthrodesis for severely destroyed joints • Joint replacement in selected patients after adequate infection control • Spinal decompression and stabilization for Potts spine when indicated **Conclusion** TB of the spine and major joints is a slowly progressive destructive disease Early diagnosis and adequate ATT can prevent deformity and neurological or functional disability --- **4 PYOGENIC**

ARTHRITIS Definition Pyogenic arthritis is an acute or chronic inflammatory infection of a synovial joint caused by pyogenic microorganisms resulting in accumulation of infected synovial fluid/pus and potential destruction of articular cartilage The terms pyogenic arthritis and septic arthritis are often used interchangeably **Etiology** Common organisms • Staphylococcus aureus most common • Streptococci • Gram negative bacilli • Neisseria gonorrhoeae in appropriate populations • Other organisms depending on age and risk factors **Routes of Infection** 1 Hematogenous Most common route 2 Direct inoculation • Penetrating injury • Joint injection • Surgery 3 Contiguous spread • Osteomyelitis • Soft tissue infection • Bursitis **Risk Factors** • Diabetes • Immunosuppression • Older age • Recent surgery • Joint prosthesis • IV drug use • Skin infection • Pre existing joint disease • Trauma **Pathophysiology** Bacterial entry into joint ↓ Synovial inflammation ↓ Neutrophil accumulation ↓ Purulent effusion ↓ Raised intra articular pressure ↓ Reduced cartilage nutrition ↓ Cartilage destruction ↓ Subchondral bone destruction ↓ Joint deformity / ankylosis **Clinical Features Acute presentation** • Severe joint pain • Rapidly developing swelling • Warmth • Redness • Tenderness • Marked restriction of movement • Severe pain on passive movement **Systemic** • Fever • Chills • Malaise • Tachycardia **Common joints** • Knee • Hip • Shoulder • Elbow • Ankle **Investigations Blood** • CBC • ESR • CRP • Blood cultures **Joint aspiration essential investigation** Synovial fluid may show • Turbid/purulent appearance • High leukocyte count • Predominantly neutrophils • Gram stain • Culture and sensitivity **Imaging** • X ray • Ultrasound • MRI where required **Complications** • Cartilage destruction • Joint space loss • Osteomyelitis • Joint deformity • Ankylosis • Sepsis • Septic shock • Pathological fracture • Growth disturbance in children • Permanent disability **Medical Management** This is an orthopedic emergency • Urgent joint aspiration • Blood cultures • Prompt empirical antibiotics after appropriate cultures then modify according to results • Analgesics • Antipyretics • Fluid management if systemically ill • Treatment of underlying conditions • Temporary immobilization in the acute painful stage **Surgical Management** Main objectives • Drain pus • Reduce bacterial load • Relieve intra articular pressure • Prevent cartilage destruction **Procedures** • Needle aspiration • Arthroscopic lavage and drainage • Open arthrotomy and drainage when indicated • Debridement • Repeat drainage if necessary **After infection is controlled** • Gradual mobilization • ROM exercises • Muscle strengthening • Functional rehabilitation **Flow Chart** Bacterial invasion of joint ↓ Synovitis ↓ Purulent effusion ↓ Raised intra articular pressure ↓ Cartilage destruction ↓ Joint destruction ↓ Early antibiotics + drainage ↓ Rehabilitation **Conclusion** Pyogenic arthritis is a joint threatening emergency Early aspiration appropriate antibiotics and adequate drainage are essential to preserve joint function --- **5 SEPTIC ARTHRITIS Definition** Septic arthritis is an infection of a joint caused by microorganisms producing inflammation and destruction of the synovium cartilage and sometimes subchondral bone Important examination point Pyogenic arthritis and septic arthritis are commonly used as synonyms Septic arthritis is the broader clinical term while pyogenic arthritis generally emphasizes bacterial suppurative infection **Etiology** Infective organisms • Staphylococcus aureus • Streptococci • Gram negative organisms • Gonococci • Other organisms in immunocompromised patients **Routes** • Hematogenous spread • Direct inoculation • Spread from adjacent osteomyelitis • Surgical contamination • Prosthetic joint infection **Risk Factors** • Diabetes mellitus • Immunosuppression • Rheumatoid arthritis • Prosthetic joint • Recent joint surgery • Skin infection • Intravenous drug use • Extremes of age • Previous septic arthritis **Clinical Features Local** • Severe joint pain • Swelling • Warmth • Redness • Tenderness • Restricted active and passive ROM • Severe pain on passive movement **Systemic** • Fever • Chills • Malaise • Tachycardia • Leukocytosis **Hip septic arthritis** May present with • Severe hip/groin pain • Refusal to bear weight • Painful hip movement • Flexion and external rotation posture in some cases **Investigations Blood** • CBC • ESR • CRP • Blood cultures **Joint aspiration Most important investigation** • Appearance • Cell count • Differential count • Gram stain • Culture and sensitivity • Crystal analysis where clinically appropriate **Imaging** • X ray • Ultrasound • MRI **Complications** • Articular cartilage destruction • Osteomyelitis • Joint deformity • Fibrous/bony ankylosis • Subluxation/dislocation • Growth plate damage • Sepsis • Septic shock • Permanent loss of joint function **Medical Management** • Urgent orthopedic evaluation • Joint aspiration • Blood cultures • Prompt IV antibiotics then culture directed therapy • Analgesia • Fluid/electrolyte management • Treatment of underlying diseases • Nutritional support **Surgical Management** Drainage methods • Repeated needle aspiration in selected joints • Arthroscopic drainage and lavage • Open arthrotomy Indications for operative drainage • Large purulent collection • Failure of aspiration • Deep joint infection • Hip infection • Inadequate clinical response • Loculated infection • Associated abscess **Rehabilitation** After infection is adequately controlled **Early** • Pain control • Gentle ROM • Positioning • Prevention of contractures **Later** • Active ROM • Isometric exercises • Progressive strengthening • Weight bearing progression • Gait training • Functional activities --- **6 SYPHILITIC INFECTION OF JOINTS Definition** Syphilitic arthritis is a rare musculoskeletal manifestation of infection with Treponema pallidum in which syphilis produces inflammatory or destructive changes involving joints bones or surrounding tissues It may occur in

different stages of syphilis particularly as a manifestation of late/congenital syphilis **Etiology** Causative organism • *Treponema pallidum* Transmission • Sexual transmission • Congenital transmission from infected mother to fetus **Types of Joint Involvement** 1 Acute syphilitic arthritis May occur during secondary syphilis 2 Chronic syphilitic arthropathy May occur in late disease 3 Charcot arthropathy A neuropathic arthropathy associated with tabes dorsalis a late neurological manifestation of syphilis 4 Congenital syphilitic skeletal disease May affect ◦ Long bones ◦ Joints ◦ Epiphyses/metaphyses **Pathophysiology** *Treponema pallidum* infection ↓ Systemic infection ↓ Inflammatory response ↓ Synovial and periarticular involvement ↓ Chronic tissue damage In late neurosyphilis Dorsal column sensory loss ↓ Loss of proprioception and protective sensation ↓ Repetitive joint trauma ↓ Progressive neuropathic arthropathy ↓ Joint destruction and deformity **Clinical Features** **Acute syphilitic arthritis** • Joint pain • Joint swelling • Stiffness • Effusion • Restricted movement • May involve multiple joints • Constitutional symptoms may occur **Chronic syphilitic arthropathy** • Chronic joint pain or discomfort • Swelling • Progressive deformity • Restricted movement **Charcot joint** Characteristic features • Recurrent painless or relatively painless trauma • Swelling • Warmth • Deformity • Crepitus • Subluxation/dislocation • Severe radiological destruction • The knee and hip can be affected among other joints **Investigations** **Serological tests** Non treponemal tests • VDRL • RPR Treponemal tests • TPHA/TPPA • FTA ABS or equivalent treponemal assays **Joint investigations** • Joint aspiration when indicated • Synovial fluid analysis • Culture to exclude pyogenic infection **Imaging** • X ray • MRI when required **Neurological evaluation** If Charcot arthropathy is suspected • Sensory examination • Proprioception • Reflexes • Neurological assessment **Differential Diagnosis** • Rheumatoid arthritis • Osteoarthritis • Septic arthritis • Tuberculous arthritis • Gout • Neuropathic arthropathy due to other causes **Complications** • Joint destruction • Deformity • Subluxation • Dislocation • Pathological fracture • Instability • Loss of function • Secondary osteoarthritis • Neurological disability in associated neurosyphilis **Medical Management** **Specific treatment** Penicillin based therapy is the standard treatment for syphilis with the exact preparation dose and duration determined by the stage and clinical form of disease Management may include • Appropriate penicillin therapy • Treatment of sexual partners according to public health guidance • Screening for other sexually transmitted infections • Neurological evaluation when indicated • Analgesics • Management of secondary complications **For Charcot arthropathy** • Protection from excessive loading • Off loading • Appropriate orthosis • Footwear modification where applicable • Management of associated neurological disease **Surgical Management** Surgery is generally directed toward the resulting structural damage rather than eradication of the infection itself May include • Synovectomy in selected cases • Debridement when indicated • Arthrodesis for severe instability • Joint reconstruction • Joint replacement in selected patients after appropriate infection treatment • Correction of severe deformity **Rehabilitation** • Protection of affected joint • Pain management • Maintaining safe ROM • Muscle strengthening • Orthotic support • Gait training • Assistive devices when required • Patient education regarding joint protection////////// **1 OSTEOMA DEFINITION** Osteoma is a benign slow growing tumour composed of mature compact or cancellous bone It commonly occurs in the skull and facial bones and rarely causes malignant transformation **CLASSIFICATION** **According to site** Central osteoma Develops within bone Peripheral osteoma Develops from the surface of bone **According to structure** Compact osteoma Cancellous osteoma **ETIOLOGY** The exact cause is uncertain Possible factors • Developmental abnormality • Genetic factors • Previous trauma • Chronic irritation • Associated syndromes particularly Gardner syndrome **CLINICAL FEATURES** Small osteomas may be asymptomatic When symptomatic • Slow growing hard swelling • Painless mass • Facial asymmetry • Headache in some cranial lesions • Sinus obstruction • Visual symptoms if orbital structures are involved • Cosmetic deformity **INVESTIGATIONS** • X ray Dense well defined bony lesion • CT scan Best for defining the size location and relationship to surrounding structures • MRI Useful when soft tissue or intracranial involvement is suspected • Biopsy Usually not required for a typical small osteoma but may be performed when diagnosis is uncertain **COMPLICATIONS** • Cosmetic deformity • Pressure on adjacent structures • Sinus obstruction • Visual impairment • Rare neurological complications • Recurrence after incomplete excision **MEDICAL MANAGEMENT** • Asymptomatic Observation Periodic clinical/imaging follow up • Symptomatic Analgesics for pain Treatment of associated sinus symptoms **SURGICAL MANAGEMENT** Indications Pain Functional impairment Significant cosmetic deformity Rapid or uncertain growth Compression of adjacent structures Procedure Complete surgical excision is the treatment of choice for symptomatic lesions **PROGNOSIS** Osteoma is benign and usually has an excellent prognosis Recurrence is uncommon after complete excision --- **2 OSTEOSARCOMA DEFINITION** Osteosarcoma is a malignant primary bone tumour in which the malignant tumour cells produce osteoid or immature bone It is the most common primary malignant bone tumour in many clinical series particularly among adolescents **CLASSIFICATION** 1 Conventional high grade osteosarcoma • Osteoblastic • Chondroblastic • Fibroblastic 2 Other types • Telangiectatic osteosarcoma • Parosteal osteosarcoma • Periosteal osteosarcoma • Secondary osteosarcoma **COMMON SITES** Usually occurs in the metaphysis of long bones Common sites • Lower end of femur • Upper end of tibia • Upper end of humerus **ETIOLOGY / RISK FACTORS** Exact cause is unknown Risk factors • Previous radiotherapy • Paget disease of bone • Hereditary retinoblastoma • Li Fraumeni syndrome • Certain

genetic abnormalities • Rapid skeletal growth may be associated with adolescent presentation **PATHOPHYSIOLOGY** Malignant transformation of osteogenic cells ↓ Rapid tumour growth ↓ Production of malignant osteoid ↓ Destruction of normal bone ↓ Extension into soft tissues ↓ Blood borne metastasis Most commonly lungs **CLINICAL FEATURES** Local • Progressive bone pain initially intermittent later persistent • Swelling and tenderness • Reduced movement of adjacent joint • Limp if lower limb is affected Advanced • Large hard mass • Venous congestion • Pathological fracture • Severe functional limitation Systemic Systemic symptoms may be absent initially **INVESTIGATIONS** • X ray May show mixed lytic and sclerotic lesion cortical destruction periosteal reaction sunburst appearance Codmans triangle • MRI Important for local tumour extent neurovascular involvement soft tissue extension surgical planning • CT chest Used to assess pulmonary metastases • Bone scan / PET CT May be used for staging • Biopsy Definitive diagnosis requires appropriate tumour biopsy planned so that the biopsy tract can be removed during definitive surgery **COMPLICATIONS** • Pathological fracture • Local tissue destruction • Neurovascular involvement • Pulmonary metastasis • Bone metastases • Recurrence • Death if untreated **MEDICAL MANAGEMENT** Osteosarcoma is usually treated using multimodal therapy • Chemotherapy Usually given before surgery neoadjuvant chemotherapy and after surgery adjuvant chemotherapy Common agents may include combinations containing High dose methotrexate Doxorubicin Cisplatin • Supportive management Analgesics Nutritional support Psychological support Management of chemotherapy complications **SURGICAL MANAGEMENT** • Limb salvage surgery Preferred when adequate oncological margins can be achieved while preserving useful limb function May involve wide excision endoprosthetic reconstruction structural bone grafting in selected cases • Amputation Considered when limb salvage cannot achieve adequate tumour control major neurovascular involvement prevents safe resection or extensive local disease makes reconstruction unsuitable **PHYSIOTHERAPY** • Preoperative Maintain ROM Maintain muscle strength Pain management Psychological preparation • Postoperative Wound and limb protection Gradual ROM Strengthening Prosthetic training if amputation is performed Gait training Functional rehabilitation **PROGNOSIS** Prognosis depends on Tumour size Site Presence of metastasis Response to chemotherapy Surgical margins --- **3 OSTEOCHONDROMA DEFINITION** Osteochondroma is a benign bone forming tumour characterized by a cartilage capped bony projection that arises from the surface of bone and typically contains cortex and medulla continuous with the underlying bone It is the most common benign bone tumour **TYPES** • Solitary osteochondroma • Multiple hereditary osteochondromas also called hereditary multiple exostoses **COMMON SITES** Usually near the metaphysis of long bones Distal femur Proximal tibia Proximal humerus **ETIOLOGY** Usually developmental Genetic mutations involving the EXT gene family are important in hereditary multiple osteochondromas **CLINICAL FEATURES** Many are asymptomatic Symptoms include • Painless bony swelling • Cosmetic deformity • Mechanical irritation • Pain due to pressure on surrounding structures • Restricted movement • Limb length discrepancy • Nerve compression • Vascular compression **INVESTIGATIONS** • X ray Characteristic bony projection from cortex medullary continuity with parent bone • CT Useful for complex anatomical areas • MRI Useful to evaluate cartilage cap neurovascular structures suspicious malignant transformation **COMPLICATIONS** • Mechanical irritation • Fracture of stalk • Nerve compression • Vascular compression • Deformity • Limb length discrepancy • Rare malignant transformation into secondary chondrosarcoma **MEDICAL MANAGEMENT** • Asymptomatic Observation Periodic review • Symptomatic Analgesics Activity modification Physiotherapy for secondary functional problems **SURGICAL MANAGEMENT** Excision is indicated for Pain Functional impairment Neurovascular compression Significant deformity Suspicion of malignant transformation Growth after skeletal maturity Complete excision usually provides excellent results --- **4 ENCHONDROMA DEFINITION** Enchondroma is a benign cartilage forming tumour that develops within the medullary cavity of bone **COMMON SITES** Small bones of hands Feet Femur Humerus Tibia **ETIOLOGY** Usually sporadic Multiple enchondromas may occur in Ollier disease and Maffucci syndrome **CLINICAL FEATURES** Often asymptomatic and discovered incidentally Symptoms include • Local swelling • Pain • Pathological fracture • Deformity • Reduced function Note Pain in an otherwise uncomplicated lesion requires evaluation for other causes including possible malignant transformation **INVESTIGATIONS** • X ray May show well defined intramedullary lucent lesion scattered calcifications cortical thinning endosteal scalloping • MRI Useful to assess cartilage characteristics soft tissue extension aggressive features • CT Useful for mineralization and cortical assessment • Biopsy Required when imaging is atypical or malignancy is suspected **COMPLICATIONS** • Pathological fracture • Deformity • Recurrence • Rare malignant transformation to chondrosarcoma • Increased risk of malignancy in some multiple lesion syndromes **MANAGEMENT** • Asymptomatic Observation Periodic clinical/imaging follow up • Symptomatic Curettage Bone grafting or suitable cavity filling when indicated Internal fixation if required for structural stability • Malignant transformation Wide surgical excision according to tumour grade and location --- **5 EWINGS SARCOMA DEFINITION** Ewing sarcoma is a malignant small round cell tumour of bone or soft tissue most commonly affecting children and adolescents It is associated with characteristic genetic alterations classically involving EWSR1 most often through an EWSR1 FLI1 fusion **COMMON SITES** Pelvis Femur Tibia Fibula Ribs Humerus Vertebrae **ETIOLOGY** The exact cause is unknown Genetic translocations are

characteristic **PATHOPHYSIOLOGY** Malignant cellular proliferation ↓ Bone marrow involvement ↓ Cortical destruction ↓ Periosteal reaction ↓ Soft tissue extension ↓ Metastasis Lungs Bone Bone marrow **CLINICAL FEATURES** • Bone pain Swelling Tenderness • Fever Malaise Weight loss • Reduced function Limp • Pathological fracture Because fever and inflammatory markers can be prominent it may initially resemble infection **INVESTIGATIONS** • Blood CBC ESR/CRP LDH may have prognostic relevance • X ray Classically permeative/moth eaten destruction lamellated or onion skin periosteal reaction • MRI Defines local extent soft tissue involvement neurovascular relationship • CT chest Pulmonary staging • PET CT / Bone marrow assessment May be used according to staging protocol • Biopsy Required for definitive diagnosis **COMPLICATIONS** • Pathological fracture Local tissue destruction • Pulmonary metastasis Bone metastasis Bone marrow involvement • Recurrence Treatment related complications **MEDICAL MANAGEMENT** Ewing sarcoma is treated with systemic chemotherapy plus local control Chemotherapy may include multi agent regimens such as Vincristine Doxorubicin Cyclophosphamide Ifosfamide Etoposide exact regimen depends on contemporary oncology protocols **SURGICAL MANAGEMENT AND RADIOTHERAPY** • Surgery Wide local excision Limb salvage surgery Reconstruction Amputation in selected cases • Radiotherapy Ewing sarcoma is relatively radiosensitive and radiotherapy may be used when surgery cannot achieve adequate margins tumour is unresectable local control is otherwise inadequate or in selected high risk situations --- **6 GIANT CELL TUMOUR OF BONE DEFINITION** Giant cell tumour GCT of bone is a generally benign but locally aggressive bone tumour composed of mononuclear stromal cells and numerous osteoclast like giant cells It can rarely metastasize particularly to the lungs **AGE** Usually occurs in young adults most commonly between skeletal maturity and approximately 40 years **COMMON SITES** Lower end of femur Upper end of tibia Distal radius Proximal humerus It characteristically involves the epiphysis extending to the subarticular bone **CLINICAL FEATURES** • Pain Progressive swelling Tenderness • Joint stiffness Reduced joint movement • Pathological fracture **INVESTIGATIONS** • X ray Typical eccentric lytic lesion expansile lesion soap bubble appearance little or no matrix mineralization extension to subchondral bone • MRI Determines soft tissue extension relationship with neurovascular structures surgical planning • Biopsy Required for definitive diagnosis • Chest imaging May be performed because rare pulmonary metastases can occur **COMPLICATIONS** • Pathological fracture • Joint destruction • Local recurrence • Secondary osteoarthritis • Rare pulmonary metastasis • Rare malignant transformation **MEDICAL MANAGEMENT** Selected cases may be treated with Denosumab particularly when surgery would cause unacceptable morbidity or in selected unresectable/recurrent disease Pain management and supportive care **SURGICAL MANAGEMENT** • Extended curettage Often combined with high speed burring local adjuvants bone graft/cement filling • Wide excision Considered for extensive soft tissue extension recurrent disease destructive lesions selected aggressive tumours • Reconstruction Bone graft Cement Prosthetic reconstruction Arthroplasty where indicated --- **7 MULTIPLE MYELOMA DEFINITION** Multiple myeloma is a malignant plasma cell disorder characterized by clonal proliferation of plasma cells in bone marrow production of a monoclonal immunoglobulin or light chain and characteristic bone destruction anemia renal dysfunction and hypercalcemia **PATHOPHYSIOLOGY** Clonal plasma cell proliferation ↓ Bone marrow infiltration ↓ Osteoclast activation + osteoblast suppression ↓ Osteolytic bone lesions ↓ Hypercalcemia + pathological fractures **COMMON SITES** Vertebrae Ribs Skull Pelvis Proximal femur Proximal humerus **CLINICAL FEATURES** Mnemonic CRAB • C HyperCalcemia Nausea Constipation Confusion Excessive thirst • R Renal impairment Reduced renal function • A Anemia Fatigue Weakness Pallor • B Bone lesions Bone pain Pathological fractures Vertebral collapse Reduced height Other features Recurrent infections Weight loss Neurological symptoms due to vertebral collapse **INVESTIGATIONS** • Blood CBC Serum calcium Renal function Total protein Serum protein electrophoresis Immunofixation Serum free light chains • Urine Urine protein electrophoresis/immunofixation Bence Jones protein/light chains • Bone marrow Bone marrow aspiration/biopsy • Imaging Low dose whole body CT MRI PET CT where appropriate **COMPLICATIONS** • Pathological fractures Vertebral collapse Spinal cord compression • Hypercalcemia Renal failure Severe anemia Recurrent infections • Amyloidosis Thrombosis **MEDICAL MANAGEMENT** Treatment is hematology/oncology directed May include • Combination anti myeloma therapy Proteasome inhibitors Immunomodulatory drugs Monoclonal antibodies Corticosteroids • Bisphosphonates or denosumab in appropriate patients • Radiotherapy for selected painful/local lesions • Autologous stem cell transplantation in eligible patients • Supportive treatment Hydration Management of hypercalcemia Renal support Treatment of anemia Infection prevention/treatment Pain management **SURGICAL MANAGEMENT** Surgery is mainly for complications Stabilization of pathological fractures Vertebral augmentation in selected cases Spinal decompression/stabilization for neurological compromise Fixation of impending or established fractures --- **8 METASTATIC BONE TUMOURS DEFINITION** Metastatic bone tumour is a malignant tumour in bone that has spread from a primary malignant tumour located elsewhere in the body It is more common than primary malignant bone tumours **COMMON PRIMARY TUMOURS** Breast Prostate Lung Kidney Thyroid **TYPES** • Osteolytic metastases Cause bone destruction Commonly Renal cell carcinoma Lung carcinoma Thyroid carcinoma • Osteoblastic metastases Cause increased bone formation/sclerosis Classically Prostate carcinoma • Mixed lesions Breast and other cancers may produce mixed patterns **COMMON SITES** Spine Pelvis Ribs Proximal femur Proximal humerus Skull **CLINICAL**

FEATURES • Persistent bone pain Night pain Back pain • Pathological fracture Reduced mobility • Weight loss Weakness • Spinal metastasis Radicular pain Weakness Sensory loss Bowel/bladder dysfunction Spinal cord

compression **INVESTIGATIONS** • Blood CBC Calcium Alkaline phosphatase Renal and liver function Disease specific tumour markers • Imaging X ray CT MRI Bone scan PET CT • Biopsy May be required if primary tumour is unknown imaging is atypical or histological confirmation is required before treatment **COMPLICATIONS** • Pathological fracture Hypercalcemia

Spinal cord compression • Severe pain Paralysis Reduced mobility Hypercalcemic crisis Death from primary malignancy **MEDICAL MANAGEMENT** Management depends on the primary cancer May include Systemic anticancer therapy Hormonal therapy Chemotherapy Targeted therapy Immunotherapy Bisphosphonates Denosumab Analgesics

Radiotherapy Particularly useful for painful metastases local tumour control spinal cord compression in selected circumstances **SURGICAL MANAGEMENT** Indications Pathological fracture Impending fracture Spinal instability Neurological compression Severe mechanical pain Procedures Internal fixation Intramedullary nailing Plate fixation

Endoprosthetic reconstruction Spinal decompression and stabilization **PART B HIP AND AVASCULAR CONDITIONS** --- **9**

PERTHES DISEASE DEFINITION Legg Calve Perthes disease is a childhood disorder characterized by idiopathic avascular necrosis of the developing femoral head followed by fragmentation reossification and remodeling **AGE** Usually 4 10 years More common in boys **ETIOLOGY** The exact cause is unknown Possible factors Temporary interruption of blood supply Genetic predisposition Coagulation abnormalities Environmental factors Repeated

microtrauma **PATHOPHYSIOLOGY** Interruption of blood supply ↓ Femoral head ischemia ↓ Bone necrosis ↓ Fragmentation ↓ Revascularization ↓ Reossification ↓ Remodeling **STAGES** 1 Initial/necrosis stage 2 Fragmentation stage 3 Reossification stage 4 Healed/remodeling stage **CLINICAL FEATURES** • Limp Hip pain Groin pain Referred knee pain •

Reduced hip ROM Restricted abduction Restricted internal rotation • Muscle wasting Trendelenburg gait **INVESTIGATIONS** • X ray May show increased density of femoral head flattening fragmentation subchondral fracture loss of sphericity • MRI Can detect early disease before X ray changes **COMPLICATIONS** • Femoral head deformity Coxa magna Coxa plana Subluxation • Leg length discrepancy Early osteoarthritis Chronic hip pain Reduced hip function **MEDICAL/CONSERVATIVE**

MANAGEMENT Goals Maintain hip mobility Maintain femoral head containment Reduce mechanical stress Prevent deformity Methods Activity modification Avoid high impact activities Analgesics/NSAIDs when appropriate Physiotherapy ROM exercises particularly abduction and internal rotation Traction in selected cases Bracing/containment treatment in selected patients **SURGICAL MANAGEMENT** Indications Depend on age femoral head involvement loss of containment progressive deformity Procedures Femoral varus osteotomy Pelvic osteotomy Other containment

procedures **PHYSIOTHERAPY** Gentle ROM Hip abduction exercises Muscle strengthening Hydrotherapy where appropriate Gait training Avoid excessive impact Maintain functional independence --- **10 SLIPPED CAPITAL FEMORAL EPIPHYSIS**

SCFE DEFINITION Slipped capital femoral epiphysis SCFE is a disorder of adolescence in which the proximal femoral epiphysis slips relative to the femoral neck through the growth plate usually producing posteromedial displacement of the epiphysis relative to the metaphysis **AGE AND RISK FACTORS** Age Commonly 10 16 years often during the adolescent growth spurt More common in boys Risk Factors Obesity Rapid growth Endocrine disorders Hypothyroidism Growth hormone abnormalities Renal disease Previous radiotherapy **CLASSIFICATION** • Stable SCFE Patient can walk with or without crutches • Unstable SCFE Patient cannot walk even with assistance Higher risk of avascular necrosis **CLINICAL**

FEATURES • Hip pain Groin pain Thigh pain Referred knee pain Limp • Reduced hip movement • Characteristic finding During passive hip flexion the hip may move into external rotation • Shortened affected limb External rotation deformity Limited internal rotation Difficulty walking **INVESTIGATIONS** • X ray AP pelvis Frog leg lateral for stable cases avoid forced positioning in suspected unstable SCFE May show widened physis displacement of epiphysis Kleins line abnormality • MRI Useful in early/pre slip cases when radiographs are inconclusive **COMPLICATIONS** • Avascular necrosis of femoral head Chondrolysis Femoroacetabular impingement • Premature osteoarthritis Limb deformity Leg length discrepancy **MEDICAL**

MANAGEMENT SCFE requires urgent orthopaedic management Until definitive treatment Stop weight bearing Bed/rest or protected mobility Analgesia Avoid forceful hip manipulation **SURGICAL MANAGEMENT** Standard treatment is in situ fixation especially for stable slips e.g. percutaneous screw fixation For selected severe/ unstable slips more extensive corrective procedures may be considered Prophylactic fixation of contralateral hip may be considered in high risk patients **PHYSIOTHERAPY** After surgical stabilization Protected weight bearing as prescribed ROM exercises as allowed Progressive strengthening Gait training Functional rehabilitation Weight management and activity modification --- **11**

AVASCULAR NECROSIS OF BONE DEFINITION Avascular necrosis AVN is the death of bone tissue caused by loss or significant reduction of its blood supply The femoral head is a common site **CAUSES** • Traumatic Femoral neck fracture Hip dislocation • Non traumatic Long term corticosteroid use Excessive alcohol use Sickle cell disease Certain autoimmune diseases Radiation Caisson disease Hypercoagulable states Some metabolic disorders Idiopathic **PATHOPHYSIOLOGY** Reduced blood supply ↓ Ischemia ↓ Death of osteocytes ↓ Structural weakening ↓ Subchondral fracture ↓ Collapse of bone ↓ Secondary osteoarthritis **CLINICAL FEATURES** • Early May be asymptomatic

Mild hip/groin pain • Progressive Deep hip pain Pain on weight bearing Pain at rest in advanced disease Restricted ROM Limp Difficulty climbing stairs Difficulty walking **INVESTIGATIONS** • X ray May be normal in early disease Later Sclerosis Cystic changes Crescent sign Femoral head collapse Secondary osteoarthritis • MRI Most sensitive investigation for early AVN • CT Useful for assessing structural collapse **COMPLICATIONS** • Femoral head collapse Secondary osteoarthritis Severe pain • Reduced mobility Leg length discrepancy Joint stiffness Disability **MEDICAL MANAGEMENT** Depends on stage Activity modification Protected weight bearing Analgesics/NSAIDs Treatment of underlying cause reduction/discontinuation of causative drugs only under medical supervision Management of associated metabolic conditions Note No medical treatment reliably reverses established femoral head collapse **SURGICAL MANAGEMENT** • Early disease Core decompression Bone grafting in selected cases Joint preserving procedures • Advanced disease Total hip arthroplasty if significant collapse and arthritis are present **PHYSIOTHERAPY** Pain relieving modalities as appropriate Protected weight bearing ROM exercises Strengthening Gait training Assistive devices Postoperative rehabilitation after arthroplasty **PART C METABOLIC BONE DISEASES** --- **12 RICKETS DEFINITION** Rickets is a disorder of the growing skeleton caused by defective mineralization of the growth plate and newly formed bone most commonly due to vitamin D deficiency or impaired vitamin D metabolism **CAUSES** • Nutritional Vitamin D deficiency Calcium deficiency Phosphate deficiency • Malabsorption Chronic gastrointestinal disease Malabsorption syndromes • Renal Chronic kidney disease Renal tubular disorders • Genetic Hereditary vitamin D dependent rickets Hypophosphatemic rickets **PATHOPHYSIOLOGY** Vitamin D / calcium / phosphate deficiency ↓ Reduced mineralization ↓ Defective growth plate mineralization ↓ Bone deformity and weakness ↓ Skeletal abnormalities **CLINICAL FEATURES** General Growth retardation Irritability Muscle weakness Delayed development Skeletal Frontal bossing Craniotables in infants Delayed closure of fontanelle Rachitic rosary Harrison sulcus Pigeon chest Widening of wrists and ankles Bow legs Knock knees Spinal deformity Dental Delayed dentition Enamel defects **INVESTIGATIONS** • Blood Serum calcium Phosphate ALP 25 hydroxyvitamin D PTH Renal function when indicated • X ray Typical metaphyseal widening cupping fraying splaying reduced bone density **COMPLICATIONS** • Skeletal deformity Growth retardation Fractures • Muscle weakness Delayed motor development Hypocalcemic seizures in severe cases Dental abnormalities **MEDICAL MANAGEMENT** • Nutritional rickets Vitamin D supplementation Adequate calcium intake Adequate phosphate where required Balanced diet Safe sunlight exposure • Specific forms Treatment directed at underlying cause e g Phosphate supplementation for hypophosphatemic disorders Active vitamin D analogues in appropriate disorders **SURGICAL MANAGEMENT** Usually unnecessary if treated early May be required for severe residual deformity or persistent angular deformity Procedures Guided growth Corrective osteotomy --- **13**

OSTEOMALACIA DEFINITION Osteomalacia is a disorder of adults characterized by defective mineralization of newly formed bone most commonly due to vitamin D deficiency or impaired vitamin D metabolism Difference Rickets affects the growing skeleton and growth plates osteomalacia affects mature bone **CAUSES** Vitamin D deficiency Calcium deficiency Malabsorption Chronic kidney disease Liver disease affecting vitamin D metabolism Anticonvulsant medications Phosphate wasting disorders **PATHOPHYSIOLOGY** Vitamin D deficiency / impaired mineral metabolism ↓ Reduced calcium/phosphate availability ↓ Defective mineralization ↓ Accumulation of osteoid ↓ Soft and weak bones **CLINICAL FEATURES** • Diffuse bone pain Bone tenderness • Proximal muscle weakness Difficulty climbing stairs Difficulty rising from a chair • Waddling gait Fractures or pseudofractures Fatigue **INVESTIGATIONS** • Blood Calcium Phosphate ALP 25 hydroxyvitamin D PTH Renal function • X ray May show reduced bone density Looser zones/pseudofractures • Bone density May show reduced BMD but does not by itself distinguish osteomalacia from osteoporosis **COMPLICATIONS** • Pathological fractures Pseudofractures Skeletal deformity • Chronic pain Muscle weakness Falls Functional disability **MEDICAL MANAGEMENT** Vitamin D replacement Calcium supplementation Correction of phosphate deficiency where applicable Treatment of underlying malabsorption/renal disease Nutritional correction **SURGICAL MANAGEMENT AND PHYSIOTHERAPY** Surgery Generally not required May be needed for severe deformity or complicated fractures Physiotherapy Muscle strengthening Balance training Gait training Fall prevention Functional training Avoid excessive loading until bone strength improves --- **14**

OSTEOPENIA DEFINITION Osteopenia is a condition in which bone mineral density is lower than normal but not low enough to meet the diagnostic criteria for osteoporosis For postmenopausal women and men aged 50 years and older a DXA T score between -1.0 and -2.5 is generally classified as osteopenia **CAUSES / RISK FACTORS** Aging Inadequate calcium intake Vitamin D deficiency Physical inactivity Smoking Excessive alcohol Low body weight Menopause Endocrine disorders Long term glucocorticoid therapy Chronic diseases **CLINICAL FEATURES** Usually asymptomatic It may be discovered through bone mineral density testing or evaluation after a fracture **INVESTIGATIONS** DXA scan Main investigation • T score -1.0 normal • T score -1.0 to -2.5 osteopenia • T score -2.5 osteoporosis In younger adults Z scores are generally more appropriate for interpretation **COMPLICATIONS** Progression to osteoporosis Increased fracture risk Vertebral fractures Hip fractures Reduced physical function **MEDICAL MANAGEMENT** Management is based on overall fracture risk not the T score alone Adequate calcium and vitamin D intake Weightbearing exercise Resistance training Smoking cessation Avoid excessive alcohol Fall prevention Treatment of secondary causes Pharmacological therapy for selected patients with sufficiently high

fracture risk **PHYSIOTHERAPY** Resistance exercises Weight bearing activities Balance training Postural exercises Gait training Fall prevention Functional strengthening --- **15 OSTEOPOROSIS DEFINITION** Osteoporosis is a systemic skeletal disorder characterized by reduced bone strength resulting from low bone mass and/ or deterioration of bone microarchitecture and consequently an increased risk of fragility fractures **TYPES** 1 Primary osteoporosis Type I Postmenopausal osteoporosis Type II Age related/senile osteoporosis 2 Secondary osteoporosis Due to Endocrine disorders Chronic kidney disease Malabsorption Rheumatological disease Medications especially glucocorticoids Prolonged immobilization Nutritional disorders **RISK FACTORS** • Non modifiable Increasing age Female sex Previous fragility fracture Family history Menopause • Modifiable Smoking Excessive alcohol Physical inactivity Poor calcium/vitamin D intake Low body weight Prolonged immobilization • Drug related Long term glucocorticoids Some anticonvulsants Certain hormonal therapies **PATHOPHYSIOLOGY** Bone resorption exceeds bone formation ↓ Loss of bone mass ↓ Trabecular thinning and cortical porosity ↓ Reduced bone strength ↓ Fragility fracture **CLINICAL FEATURES** Often called a silent disease because bone loss may occur without symptoms until fracture develops Manifestations Back pain Vertebral compression fracture Loss of height Kyphosis Fragility fracture Hip Wrist Vertebrae Difficulty walking Reduced independence **INVESTIGATIONS** • DXA scan Standard test for bone mineral density T score thresholds apply -1 0 Normal -1 0 to -2 5 Osteopenia -2 5 Osteoporosis • Blood investigations Calcium Phosphate ALP Vitamin D Renal function Thyroid studies to identify secondary causes • X ray Fractures Vertebral compression Kyphosis **COMPLICATIONS** Vertebral compression fracture Hip fracture Wrist fracture Loss of height Kyphosis Chronic pain Reduced mobility Falls Loss of independence Increased mortality following major hip fracture **MEDICAL MANAGEMENT** 1 Lifestyle management Adequate calcium and vitamin D Weight bearing and resistance exercise Balance training Smoking cessation Limit excessive alcohol Fall prevention 2 Pharmacological treatment Bisphosphonates Denosumab Selective estrogen receptor modulators Anabolic agents teriparatide/abaloparatide Romosozumab Choice depends on fracture risk age renal function previous fractures contraindications **SURGICAL MANAGEMENT** Osteoporosis itself is usually managed medically Surgery is required for complications e.g fracture management via internal fixation arthroplasty vertebral augmentation for vertebral compression hemiarthroplasty/THA for hip fractures **PHYSIOTHERAPY MANAGEMENT** Goals Improve muscle strength Improve balance Prevent falls Maintain posture Improve functional independence Reduce fracture risk Exercises Weight bearing exercise Resistance training Back extensor strengthening Balance training Postural exercises Gait training Precautions Avoid high impact activity in patients at high fracture risk Forceful spinal flexion Excessive twisting Unsupervised heavy lifting //

/// **1 OSTEOARTHRITIS DEFINITION** Osteoarthritis OA is a chronic progressive degenerative disorder of synovial joints characterized by loss of articular cartilage subchondral bone changes osteophyte formation and varying degrees of synovial inflammation resulting in pain stiffness and functional limitation **CAUSES / RISK FACTORS** **Primary OA** Occurs without a single identifiable cause Risk factors • Increasing age • Female sex • Obesity • Genetic predisposition • Previous joint injury • Repetitive joint loading **Secondary OA** Occurs due to an identifiable condition • Previous fracture • Joint infection • Rheumatoid arthritis • Congenital joint abnormality • Ligament injury • Meniscal injury • Avascular necrosis • Metabolic disorders **COMMON SITES** • Knee • Hip • Spine • Hands • First MTP joint • Shoulder and other joints less commonly **PATHOPHYSIOLOGY** Mechanical/biological stress ↓ Cartilage matrix degeneration ↓ Loss of proteoglycans and collagen integrity ↓ Cartilage thinning and fissuring ↓ Subchondral sclerosis ↓ Osteophyte formation ↓ Joint space narrowing ↓ Pain + stiffness + deformity **CLINICAL FEATURES** **Symptoms** • Joint pain usually worse with activity • Pain relieved by rest initially • Morning stiffness usually less than 30 minutes • Stiffness after prolonged inactivity • Difficulty walking • Difficulty climbing stairs • Reduced functional activity **Signs** • Joint tenderness • Crepitus • Reduced ROM • Bony enlargement • Joint line tenderness • Muscle weakness • Deformity in advanced disease **Hand OA** • Heberdens nodes DIP joints • Bouchards nodes PIP joints **DEFORMITIES** **Knee** • Varus deformity • Valgus deformity **Hip** • Flexion deformity • External rotation • Adduction deformity **Hands** • Heberdens and Bouchards nodes **RADIOLOGICAL FEATURES** X ray shows • Non uniform joint space narrowing • Osteophytes • Subchondral sclerosis • Subchondral cysts • Bony remodeling • Deformity in advanced disease Classic mnemonic LOSS • Loss of joint space • Osteophytes • Subchondral sclerosis • Subchondral cysts **COMPLICATIONS** • Severe pain • Joint deformity • Contractures • Muscle wasting • Reduced mobility • Disability • Secondary instability • Reduced quality of life **CONSERVATIVE MANAGEMENT** **Medical** • Patient education • Weight reduction • Activity modification • Paracetamol/analgesics where appropriate • NSAIDs when appropriate • Topical NSAIDs • Intra articular corticosteroid injection in selected cases • Management of comorbidities **Physiotherapy** • ROM exercises • Quadriceps strengthening • Hip strengthening • Aerobic exercise • Aquatic exercise • Gait training • Balance exercises • Assistive devices • Bracing when indicated **SURGICAL MANAGEMENT** Indicated in severe symptomatic disease with significant functional limitation despite conservative treatment **Procedures** • Arthroscopic procedures in selected circumstances • Osteotomy in selected younger patients with appropriate deformity • Total knee replacement • Total hip replacement • Other joint replacement procedures --- **2 RHEUMATOID ARTHRITIS DEFINITION** Rheumatoid arthritis RA is a chronic systemic autoimmune inflammatory disease characterized primarily by persistent inflammatory synovitis of

multiple joints leading to cartilage destruction bone erosion deformity and extra articular manifestations **CAUSES / RISK FACTORS** Exact cause is unknown Important factors • Genetic predisposition • Autoimmune mechanisms • Female sex • Smoking • Environmental factors • Hormonal factors • Certain infections may act as triggers **PATHOPHYSIOLOGY** Autoimmune activation ↓ Synovial inflammation ↓ Pannus formation ↓ Cartilage destruction ↓ Marginal bone erosions ↓ Ligament/tendon damage ↓ Joint instability and deformity **CLINICAL FEATURES General** • Fatigue • Malaise • Fever • Loss of appetite • Weight loss **Joint** • Symmetrical polyarthritis • Pain • Swelling • Warmth • Prolonged morning stiffness typically 30 60 minutes • Reduced ROM Commonly affects • MCP • PIP • Wrist • MTP • Knee • Elbow • Shoulder • DIP joints are usually spared **CHARACTERISTIC DEFORMITIES Hand** • Ulnar deviation • Swan neck deformity • Boutonniere deformity • Z thumb deformity • Wrist deformity **Foot** • Hallux valgus • MTP subluxation • Claw toes **Cervical spine** • Atlantoaxial instability/subluxation **COMPLICATIONS Joint** • Destruction • Subluxation • Ankylosis • Deformity • Disability **Extra articular** • Rheumatoid nodules • Vasculitis • Interstitial lung disease • Pleural disease • Pericarditis • Eye involvement • Anemia • Osteoporosis • Increased cardiovascular risk **RADIOLOGICAL FEATURES Early** • Soft tissue swelling • Periarticular osteopenia **Later** • Uniform joint space narrowing • Marginal erosions • Subluxation • Deformity • Ankylosis in advanced disease **Cervical spine** • Atlantoaxial instability • Erosive changes **CONSERVATIVE MANAGEMENT Medical** Early treatment with DMARDs is central May include • Methotrexate • Hydroxychloroquine • Sulfasalazine • Leflunomide • Biological/targeted DMARDs • NSAIDs • Corticosteroids for selected indications **Physiotherapy** • Pain relief • ROM exercises • Muscle strengthening • Joint protection • Splinting • Energy conservation • Hydrotherapy • Functional training • Assistive devices **SURGICAL MANAGEMENT Indications** • Severe joint destruction • Persistent pain • Functional disability • Deformity Procedures • Synovectomy • Tendon repair • Arthrodesis • Joint replacement • Cervical stabilization when required --- **3 ANKYLOSING SPONDYLITIS DEFINITION** Ankylosing spondylitis AS is a chronic inflammatory rheumatic disease predominantly affecting the axial skeleton and sacroiliac joints causing inflammatory back pain progressive spinal stiffness and in advanced disease spinal fusion It belongs to the axial spondyloarthritis spectrum **CAUSES / RISK FACTORS** • Strong genetic association with HLA B27 • Family history • Immune mediated mechanisms • Environmental factors **CLINICAL FEATURES Early** • Insidious low back pain • Buttock pain • Sacroiliac pain • Morning stiffness • Improvement with exercise • Worsening with rest • Night pain **Progressive** • Reduced spinal mobility • Chest expansion • Hip involvement • Fatigue **DEFORMITIES** Advanced disease may produce • Loss of lumbar lordosis • Increased thoracic kyphosis • Forward head posture • Hip flexion deformity • Fixed spinal posture Classically described as a bamboo spine in advanced radiographic disease **COMPLICATIONS** • Severe spinal stiffness • Hip arthritis • Osteoporosis • Vertebral fractures • Spinal cord injury after fracture • Uveitis • Cardiovascular involvement • Restrictive chest wall mechanics • Functional disability **RADIOLOGICAL FEATURES X ray** • Sacroiliitis • Vertebral squaring • Vertebral corner erosions • Syndesmophytes • Bridging of vertebrae • Bamboo spine • Hip joint changes **MRI** Can demonstrate • Active sacroiliitis • Bone marrow edema • Early inflammatory changes **CONSERVATIVE MANAGEMENT Medical** • NSAIDs • Biologic agents such as TNF inhibitors or IL 17 inhibitors when indicated • Other targeted therapy in selected patients **Physiotherapy** Very important • Spinal mobility exercises • Postural correction • Chest expansion exercises • Breathing exercises • Strengthening • Hydrotherapy • Stretching • Aerobic conditioning • Patient education • Maintain upright posture • Avoid prolonged flexed posture • Regular exercise **SURGICAL MANAGEMENT Indications** • Severe hip destruction • Fixed severe deformity • Neurological compression • Unstable spinal fracture Procedures • Total hip replacement • Corrective spinal osteotomy in selected severe deformities • Spinal stabilization/decompression when indicated --- **4 GOUTY ARTHRITIS DEFINITION** Gout is an inflammatory arthritis caused by deposition of monosodium urate crystals in joints and tissues as a result of persistent hyperuricemia **CAUSES / RISK FACTORS Primary** Usually related to impaired uric acid excretion or increased production **Secondary** • Chronic kidney disease • Myeloproliferative disorders • Certain medications • Tumour lysis • High cell turnover Risk factors • Obesity • Alcohol excess • High purine diet • Metabolic syndrome • Renal disease • Family history **COMMON SITES** • First MTP joint • Podagra • Ankle • Knee • Midfoot • Elbow • Wrist • Fingers **CLINICAL FEATURES Acute attack** • Sudden severe pain • Redness • Swelling • Warmth • Extreme tenderness • Often occurs at night **Chronic gout** • Recurrent attacks • Tophi • Joint destruction • Chronic stiffness **Tophi** Deposits may occur in • Ear • Fingers • Toes • Elbow • Achilles tendon **COMPLICATIONS** • Chronic arthritis • Tophi • Joint destruction • Kidney stones • Chronic kidney disease • Reduced function **DEFORMITIES** • Irregular joint enlargement • Destructive deformity • Tophi related deformity **RADIOLOGICAL FEATURES X ray** • Well defined punched out erosions • Overhanging edges • Preserved joint space until later disease • Soft tissue tophi • Bone destruction in advanced disease **CONSERVATIVE MANAGEMENT Acute attack** • NSAIDs • Colchicine • Corticosteroids • Rest and ice • Avoid joint trauma **Long term** • Weight management • Reduce alcohol • Dietary modification • Adequate hydration • Urate lowering therapy • Allopurinol • Febuxostat in appropriate patients • Treatment should be individualized based on renal function and comorbidities **SURGICAL MANAGEMENT** Usually not required May be indicated for • Large symptomatic tophi • Nerve compression • Severe joint destruction • Recurrent infection over tophi •

Severe functional impairment Procedures • Tophus excision • Joint reconstruction/replacement in selected cases --- **5**

PSORIATIC ARTHRITIS DEFINITION Psoriatic arthritis PsA is a chronic inflammatory musculoskeletal disease associated with psoriasis involving peripheral joints the axial skeleton entheses and digits **CAUSES / RISK FACTORS** • Genetic predisposition

• Psoriasis • Immune dysregulation • Family history • Environmental factors **CLINICAL FEATURES Joint** • Asymmetrical arthritis • Pain • Swelling • Stiffness • Morning stiffness **Characteristic** • Dactylitis sausage digit • Enthesitis • Commonly •

Achilles tendon • Plantar fascia **Skin** • Psoriatic plaques • Nail pitting • Nail dystrophy **Spine** • Inflammatory back pain in axial disease **DEFORMITIES** • Joint deformity • Flexion deformities • Pencil in cup deformity in severe disease • Arthritis mutilans • Ankylosis **COMPLICATIONS** • Permanent joint damage • Severe functional disability • Spinal stiffness • Eye

inflammation • Cardiovascular/metabolic comorbidity **RADIOLOGICAL FEATURES** • Erosions • New bone formation • Asymmetric joint space loss • Periostitis • Pencil in cup deformity • Ankylosis • Sacroiliitis •

Syndesmophytes **CONSERVATIVE MANAGEMENT Medical** • NSAIDs • DMARDs • Methotrexate • Sulfasalazine in selected peripheral disease • Biologic agents • Targeted synthetic DMARDs • Treatment of psoriasis **Physiotherapy** • ROM exercises

• Strengthening • Hydrotherapy • Postural exercises • Joint protection • Functional training **SURGICAL MANAGEMENT** For advanced joint destruction • Synovectomy in selected cases • Arthrodesis • Joint replacement • Tendon procedures --- **6**

HAEMOPHILIC ARTHRITIS DEFINITION Haemophilic arthritis is a chronic destructive arthropathy caused by recurrent bleeding into joints hemarthrosis in patients with haemophilia It commonly affects • Knee • Ankle • Elbow **CAUSE** Usually occurs in • Hemophilia A factor VIII deficiency • Hemophilia B factor IX deficiency Repeated hemarthroses produce chronic synovial inflammation and cartilage destruction **PATHOPHYSIOLOGY** Factor deficiency ↓ Recurrent hemarthrosis ↓ Blood

accumulation in joint ↓ Synovial hypertrophy ↓ Iron deposition ↓ Cartilage destruction ↓ Joint space narrowing ↓ Deformity **CLINICAL FEATURES Acute hemarthrosis** • Sudden joint pain • Swelling • Warmth • Restricted movement •

Protective muscle spasm **Chronic** • Recurrent swelling • Chronic pain • Stiffness • Muscle wasting • Flexion contractures • Reduced ROM **DEFORMITIES** • Knee flexion deformity • Elbow flexion deformity • Ankle equinus • Joint enlargement •

Muscle wasting **COMPLICATIONS** • Chronic arthropathy • Contractures • Joint deformity • Muscle wasting • Secondary osteoarthritis • Severe disability • Recurrent bleeding **RADIOLOGICAL FEATURES** Early • Soft tissue swelling • Joint effusion

Advanced • Epiphyseal enlargement • Subchondral cysts • Joint space narrowing • Osteophytes • Erosions • Irregular articular surfaces MRI can identify • Synovial hypertrophy • Hemosiderin deposition • Cartilage damage **CONSERVATIVE**

MANAGEMENT Acute bleed Factor replacement is the main treatment Additional • Rest • Ice • Compression where appropriate • Elevation • Analgesia • Protected movement initially **Long term** • Factor prophylaxis where

indicated **Physiotherapy** • Muscle strengthening • ROM exercises • Joint protection • Weight management **SURGICAL**

MANAGEMENT For advanced arthropathy • Synovectomy • Joint replacement • Arthroscopic procedures in selected cases • Corrective procedures for severe deformity Adequate factor replacement is essential around surgery --- **7 STILLS DISEASE**

/ JUVENILE IDIOPATHIC ARTHRITIS DEFINITION Systemic juvenile idiopathic arthritis Stills disease is a subtype of juvenile idiopathic arthritis characterized by arthritis with systemic inflammation fever and often an evanescent salmon pink rash **CAUSES** Exact cause is unknown Possible mechanisms • Autoinflammatory immune dysfunction • Genetic

predisposition • Environmental triggers **CLINICAL FEATURES Systemic** • High spiking fever • Evanescent salmon pink rash • Fatigue • Malaise • Lymphadenopathy • Hepatosplenomegaly • Serositis **Joint** • Pain • Swelling • Stiffness • Reduced ROM

• Polyarthritis **DEFORMITIES** Untreated chronic disease may produce • Flexion deformities • Growth disturbance • Limb length discrepancy • Joint destruction • Contractures **COMPLICATIONS** • Growth retardation • Joint destruction •

Contractures • Osteoporosis • Macrophage activation syndrome • Chronic disability • Amyloidosis in longstanding inflammatory disease **RADIOLOGICAL FEATURES** Early • Soft tissue swelling • Osteopenia Later • Joint space narrowing •

Growth abnormalities • Erosions • Deformity **CONSERVATIVE / MEDICAL MANAGEMENT** • NSAIDs • Corticosteroids • Methotrexate for persistent arthritis • Biologic agents especially IL 1 or IL 6 targeted therapy where indicated •

Physiotherapy • Occupational therapy • Nutritional support **SURGICAL MANAGEMENT** Rarely required in active systemic disease In severe chronic joint destruction • Synovectomy in selected cases • Joint replacement • Corrective surgery --- **8**

CHARCOT JOINT DEFINITION Charcot joint neuropathic arthropathy is a progressive destructive arthropathy resulting from loss of protective sensation and proprioception leading to repetitive microtrauma joint instability deformity and

destruction **CAUSES** Most common • Diabetes mellitus with peripheral neuropathy Other causes • Tabes dorsalis • Syringomyelia • Peripheral neuropathy • Spinal cord disease • Congenital sensory neuropathy **COMMON SITES** • Foot and

ankle • Midfoot • Knee • Hip • Spine **PATHOPHYSIOLOGY** Loss of protective sensation ↓ Repeated unrecognized trauma ↓ Inflammation + bone resorption ↓ Joint instability ↓ Fragmentation and dislocation ↓ Severe deformity **CLINICAL**

FEATURES • Swelling • Warmth • Redness • Instability • Deformity • Reduced pain despite severe structural damage • Crepitus • Altered gait **Foot** • Characteristic rocker bottom deformity may occur **COMPLICATIONS** • Severe deformity •

Ulceration • Infection • Osteomyelitis • Amputation • Fractures • Instability • Loss of mobility **RADIOLOGICAL**

FEATURES Early • Soft tissue swelling • Subchondral changes Advanced • Joint destruction • Fragmentation • Subluxation •

Dislocation • Bone sclerosis • Osteophytes • Debris • Deformity A classic description is bag of bones appearance in advanced disease **CONSERVATIVE MANAGEMENT Acute stage** • Complete/off loading as indicated • Total contact casting or appropriate immobilization • Custom orthoses • Regular skin inspection • Glycemic control • Foot care education **Chronic** • Custom footwear • Orthoses • Protected weight bearing • Gait training • Fall prevention **SURGICAL MANAGEMENT** Indications • Severe deformity • Instability • Recurrent ulceration • Failure of conservative treatment • Severe structural destruction Procedures • Arthrodesis • Corrective osteotomy • Exostectomy in selected cases • Reconstruction • Amputation for uncontrollable infection or non reconstructable limb **CONNECTIVE TISSUE DISORDERS --**

- 9 SYSTEMIC LUPUS ERYTHEMATOSUS SLE DEFINITION Systemic lupus erythematosus SLE is a chronic systemic autoimmune disease characterized by autoantibody production and immune mediated inflammation affecting multiple organs including the skin joints kidneys nervous system blood and cardiovascular system **CAUSES / RISK FACTORS** • Genetic predisposition • Female sex • Hormonal factors • Environmental triggers • Ultraviolet light • Certain medications • Infections may act as triggers **CLINICAL FEATURES General** • Fatigue • Fever • Weight loss **Musculoskeletal** • Symmetrical joint pain • Non erosive inflammatory arthritis • Morning stiffness • Myalgia **Skin** • Malar rash • Photosensitivity • Discoid lesions **Other systems** • Oral ulcers • Hair loss • Renal disease • Neurological manifestations • Serositis • Hematological abnormalities **DEFORMITIES** Usually non erosive arthritis but chronic disease may cause • Jaccoud arthropathy • Ulnar deviation • Swan neck like deformities • Tendon abnormalities These may be partially reducible unlike fixed erosive RA deformities **COMPLICATIONS** • Lupus nephritis • Cardiovascular disease • Stroke • Seizures • Anemia • Thrombocytopenia • Osteoporosis • Avascular necrosis particularly with prolonged corticosteroid exposure • Infection **RADIOLOGICAL FEATURES** Usually • Soft tissue swelling • Osteopenia may occur Unlike RA • Marked erosions are usually absent X rays may show • Jaccoud type deformities without typical erosive destruction **CONSERVATIVE / MEDICAL MANAGEMENT** • Patient education • Sun protection • Exercise • Hydroxychloroquine in most appropriate patients • NSAIDs for selected musculoskeletal symptoms • Corticosteroids for active disease when indicated • Immunosuppressive therapy • Biologic therapy in selected patients • Treatment of individual organ involvement **SURGICAL MANAGEMENT** Not usually directed at SLE itself May be required for • Severe joint destruction • Avascular necrosis • Fracture • Tendon deformity Procedures • Joint replacement • Corrective surgery • Fracture fixation --- **10 SCLERODERMA / SYSTEMIC SCLEROSIS DEFINITION** Systemic sclerosis scleroderma is a chronic autoimmune connective tissue disease characterized by vascular abnormalities immune dysregulation and excessive collagen deposition resulting in fibrosis and hardening of skin and potentially internal organs **TYPES** 1 Limited cutaneous systemic sclerosis Skin involvement mainly distal to elbows/knees and face 2 Diffuse cutaneous systemic sclerosis More extensive skin involvement including trunk and proximal limbs **CAUSES / RISK FACTORS** Exact cause unknown Factors • Genetic predisposition • Autoimmune mechanisms • Environmental exposures • Vascular dysfunction **CLINICAL FEATURES Skin** • Skin thickening • Tight shiny skin • Loss of skin folds • Reduced mobility **Hands** • Raynaud phenomenon • Puffy fingers initially • Sclerodactyly • Digital ulcers • Contractures **Face** • Tight facial skin • Reduced mouth opening • Mask like appearance **Systemic** • Dysphagia • Reflux • Interstitial lung disease • Pulmonary hypertension • Renal crisis **DEFORMITIES** • Flexion contractures of fingers • Sclerodactyly • Reduced mouth opening • Hand deformity • Joint stiffness **COMPLICATIONS** • Digital ulcers • Gangrene • Pulmonary fibrosis • Pulmonary hypertension • Renal crisis • Cardiac disease • Esophageal dysfunction • Severe functional disability **RADIOLOGICAL FEATURES** May show • Soft tissue calcification • Acro osteolysis • Joint space changes • Contractures Chest imaging may demonstrate • Interstitial lung disease/fibrosis **CONSERVATIVE MANAGEMENT Medical** • Vasodilator therapy for Raynaud phenomenon • Immunosuppressive therapy for selected systemic disease • Proton pump inhibitors for reflux • Treatment of pulmonary hypertension • Renal blood pressure management **Physiotherapy** • ROM exercises • Stretching • Hand exercises • Splinting when indicated • Breathing exercises • Aerobic conditioning • Functional training • Skin protection and education **SURGICAL MANAGEMENT** May include • Debridement of severe digital ulcers • Management of gangrene • Joint reconstruction in selected cases • Surgical treatment of severe contractures --- **11 DERMATOMYOSITIS DEFINITION** Dermatomyositis is an autoimmune inflammatory disease characterized by proximal skeletal muscle weakness and characteristic cutaneous manifestations **CAUSES / RISK FACTORS** Exact cause unknown Possible associations • Autoimmune predisposition • Genetic factors • Environmental triggers • In some adults association with malignancy **CLINICAL FEATURES Muscle** • Symmetrical proximal muscle weakness • Difficulty climbing stairs • Difficulty rising from a chair • Difficulty lifting arms • Difficulty combing hair • Neck flexor weakness **Skin** • Heliotrope rash • Gottron papules • Photosensitive rash • Shawl sign • V sign **Other** • Dysphagia • Respiratory muscle involvement • Fatigue **DEFORMITIES / FUNCTIONAL PROBLEMS** • Flexion contractures • Poor posture • Muscle wasting • Reduced mobility • Difficulty with transfers • Gait abnormalities • Children may develop calcinosis **COMPLICATIONS** • Severe muscle weakness • Dysphagia • Aspiration • Respiratory failure • Interstitial lung disease • Calcinosis • Malignancy association in some adults • Disability **RADIOLOGICAL FEATURES** There are no characteristic plain X ray findings MRI muscle may show • Muscle edema • Inflammatory changes • Muscle atrophy in chronic disease Other investigations • Elevated CK • EMG

abnormalities • Muscle biopsy in selected cases **CONSERVATIVE / MEDICAL MANAGEMENT** Medical • Corticosteroids • Immunosuppressive agents • IVIG in selected cases • Other immunomodulatory/biologic therapies as appropriate • Sun protection • Treatment of dysphagia and respiratory complications **Physiotherapy** • Gentle ROM • Gradual strengthening after disease control • Aerobic conditioning • Functional training • Energy conservation • Respiratory exercises where indicated • Avoid aggressive exercise during uncontrolled active myositis **SURGICAL MANAGEMENT** Usually not required for the primary disease May be required for • Severe calcinosis • Contractures • Secondary complications --- **12 POLIOMYELITIS Important** Poliomyelitis is primarily an infectious viral disease of the nervous system not a connective tissue disease It is included here because it is part of the requested syllabus **DEFINITION** Poliomyelitis is an acute viral infection caused by poliovirus primarily affecting the anterior horn cells of the spinal cord and sometimes the brainstem resulting in flaccid muscle weakness or paralysis **CAUSE** Causative organism • Poliovirus Transmission • Mainly fecal oral route • Occasionally oral oral transmission **PATHOPHYSIOLOGY** Poliovirus infection ↓ Viral replication ↓ CNS invasion in severe disease ↓ Anterior horn cell destruction ↓ Lower motor neuron dysfunction ↓ Flaccid paralysis **CLINICAL FEATURES** **Prodromal** • Fever • Headache • Malaise • Sore throat • Vomiting **Paralytic disease** • Acute asymmetric weakness • Flaccid paralysis • Reduced muscle tone • Reduced/absent reflexes • Muscle wasting • Fasciculations may occur • Sensory function usually preserved **Respiratory involvement** • Intercostal weakness • Diaphragmatic weakness • Respiratory failure **DEFORMITIES** Long term sequelae may include • Equinus deformity • Knee flexion deformity • Hip deformity • Scoliosis • Limb shortening • Foot deformity • Joint contractures **COMPLICATIONS** • Respiratory failure • Aspiration • Permanent paralysis • Muscle wasting • Contractures • Skeletal deformity • Post polio syndrome years later • Reduced functional independence **RADIOLOGICAL FEATURES** X rays are not diagnostic of acute poliomyelitis Chronic cases may show • Muscle wasting related osteopenia • Limb shortening • Deformities • Scoliosis • Joint contractures **CONSERVATIVE MANAGEMENT** **Acute** • Supportive medical care • Respiratory support • Pain control • Prevention of contractures • Appropriate positioning **Rehabilitation** • ROM exercises • Muscle strengthening of functioning muscles • Orthoses • Gait training • Energy conservation • Assistive devices • Respiratory physiotherapy where indicated **Post polio syndrome** • Pacing • Energy conservation • Avoid overuse weakness • Appropriate low to moderate exercise • Functional adaptation **SURGICAL MANAGEMENT** For residual deformities • Tendon transfer • Corrective osteotomy • Arthrodesis • Limb length procedures in selected cases • Spinal surgery for severe scoliosis • Orthopaedic reconstruction --- **13 MIXED CONNECTIVE TISSUE DISEASE MCTD DEFINITION** Mixed connective tissue disease MCTD is an autoimmune connective tissue disorder characterized by overlapping clinical features of systemic lupus erythematosus systemic sclerosis polymyositis/dermatomyositis and rheumatoid arthritis usually associated with high levels of anti U1 RNP antibodies **CAUSES** Exact cause is unknown Associated with • Genetic predisposition • Autoimmune dysregulation • Environmental triggers **CLINICAL FEATURES** **Musculoskeletal** • Polyarthritides • Arthralgia • Myalgia • Proximal muscle weakness **Vascular** • Raynaud phenomenon **Skin** • Puffy fingers • Sclerodactyly like changes • Rash **Other** • Dysphagia • Esophageal dysfunction • Interstitial lung disease • Pulmonary hypertension • Fatigue **DEFORMITIES** May develop • Joint deformity • Contractures • Sclerodactyly • Muscle wasting • Reduced hand function **COMPLICATIONS** • Pulmonary hypertension • Interstitial lung disease • Esophageal dysfunction • Cardiac involvement • Renal involvement usually less prominent than in classic SLE • Severe muscle weakness • Functional disability **RADIOLOGICAL FEATURES** No single pathognomonic radiographic appearance May demonstrate • Joint space narrowing • Soft tissue swelling • Erosive arthritis in some patients • Interstitial lung disease on chest X ray/HRCT • Muscle inflammation/edema on MRI **INVESTIGATIONS** **Blood** • ANA • Anti U1 RNP • CBC • ESR/CRP • CK • Renal and liver function **Other** • Pulmonary function tests • Echocardiography • HRCT chest where indicated • MRI muscle if myositis is suspected **CONSERVATIVE / MEDICAL MANAGEMENT** Treatment depends on the dominant manifestations May include • NSAIDs • Corticosteroids • Hydroxychloroquine • Methotrexate • Other immunosuppressive drugs • Vasodilators for Raynaud phenomenon • Treatment for pulmonary hypertension • Treatment of interstitial lung disease **Physiotherapy** • ROM exercises • Muscle strengthening • Aerobic conditioning • Breathing exercises • Hand therapy • Joint protection • Energy conservation • Functional training **SURGICAL MANAGEMENT** Usually not required for the primary disease May be considered for • Severe joint destruction • Fixed deformity • Contractures • Digital complications • Joint replacement in selected cases

CO 5 -- 1 CERVICOBRACHIAL SYNDROME Definition Cervicobrachial syndrome is a clinical condition characterized by pain paraesthesia numbness weakness or other neurological symptoms originating from the cervical spine or surrounding structures and radiating into the shoulder arm forearm or hand **Causes / Etiology** • Cervical intervertebral disc prolapse • Cervical spondylosis and osteophytes • Foraminal or spinal canal stenosis • Whiplash and other cervical trauma • Muscle spasm and postural dysfunction • Brachial plexus lesions • Tumour infection or inflammatory disorders • Thoracic outlet disorders **Pathophysiology** Cervical pathology → nerve root irritation/compression → inflammation and impaired nerve conduction → pain and paraesthesia → weakness and functional limitation **Clinical Features** • Neck pain with radiation to shoulder/upper limb • Burning shooting or electric shock like pain • Paraesthesia and numbness in a dermatomal pattern •

Muscle weakness and reduced grip strength • Reduced deep tendon reflexes corresponding to the affected root • Restricted cervical ROM and muscle spasm • Symptoms may worsen with cervical movement or compression **Investigations** • Cervical X ray for degenerative or bony changes • MRI for disc pathology nerve root or cord compression • CT for bony/foraminal abnormalities • EMG/NCV when nerve involvement needs clarification • Neurological examination of dermatomes myotomes and reflexes **Complications** • Chronic pain • Persistent sensory deficit • Motor weakness and muscle wasting • Loss of hand function • Reduced work capacity • Cervical myelopathy in severe cord compression **Conservative / Medical Management** • Relative rest and activity modification • Analgesics/NSAIDs as indicated • Muscle relaxants in selected cases • Neuropathic pain medication when appropriate • Selected use of cervical collar in acute cases **Physiotherapy Management** • Pain relief with TENS or thermal modalities when appropriate • Gentle cervical ROM • Deep neck flexor and cervical stabilization exercises • Scapular stabilization and upper limb strengthening • Postural and ergonomic correction • Cervical traction in selected patients without contraindications • Neural mobilization when appropriate • Home exercise programme **Surgical Management** • Indications progressive neurological deficit severe persistent radicular pain despite appropriate conservative treatment significant structural compression myelopathy or instability • Anterior cervical discectomy and fusion ACDF • Cervical disc replacement in selected patients • Posterior cervical foraminotomy • Cervical decompression/laminectomy or laminoplasty when indicated **Conclusion** Early identification of neurological deficits appropriate investigation posture and exercise based rehabilitation and timely surgical referral in progressive or severe cases are essential ---

2 THORACIC OUTLET SYNDROME Definition Thoracic outlet syndrome TOS is a group of disorders caused by compression of the brachial plexus and/or subclavian vessels as they pass through the thoracic outlet producing neurological or vascular symptoms in the upper limb **Types** • Neurogenic TOS brachial plexus involvement • Venous TOS subclavian vein involvement • Arterial TOS subclavian artery involvement **Causes / Risk Factors** • Cervical rib or first rib abnormality • Fibrous bands or abnormal scalene muscles • Forward head and rounded shoulder posture • Clavicular/first rib trauma • Repetitive overhead activity • Heavy lifting and occupational overuse **Pathophysiology** Anatomical abnormality or postural dysfunction → narrowing of thoracic outlet → neurovascular compression → neurological and/or vascular symptoms **Clinical Features** • Neck shoulder and arm pain • Paraesthesia numbness and weak grip • Symptoms aggravated by overhead activity • Venous type swelling cyanosis heaviness and superficial veins • Arterial type coldness pallor reduced pulse and ischaemic symptoms • Possible positive Adson Roos/EAST Wright or costoclavicular manoeuvres **Investigations** • X ray of cervical spine/chest for cervical rib • Doppler ultrasound • CT angiography or MR angiography for vascular disease • MRI when structural pathology is suspected • EMG/NCV in selected neurogenic cases **Complications** • Chronic neuropathic pain • Muscle weakness and wasting • Venous thrombosis • Arterial thrombosis or distal embolization • Limb ischaemia • Functional disability **Conservative / Physiotherapy Management** • Activity modification • Postural correction • Scalene and pectoralis minor stretching • Thoracic mobility exercises • Deep neck flexor strengthening • Scapular stabilization • Breathing exercises • Ergonomic correction • Neural mobilization when appropriate • Avoid prolonged overhead activity and heavy loads on the affected side **Surgical Management** • Indications include severe vascular compromise thrombosis progressive neurological deficit significant structural compression or failure of appropriate conservative care • First rib resection • Scalenectomy/scalenotomy • Cervical rib excision • Fibrous band release • Vascular reconstruction when required **Conclusion** Management depends on whether the syndrome is neurological venous or arterial rehabilitation is important in appropriate cases while vascular emergencies require urgent medical treatment ---

3 VERTEBROBASILAR SYNDROME Definition Vertebrobasilar syndrome refers to neurological manifestations caused by reduced blood flow or ischaemia in the vertebral and basilar arterial circulation supplying the brainstem cerebellum and posterior cerebral structures **Causes** • Atherosclerosis • Thrombosis or embolism • Vertebral artery dissection • Hypertension diabetes and hyperlipidaemia • Cervical arterial injury or trauma • Rare vascular abnormalities or compression **Pathophysiology** Vascular disease → reduced vertebrobasilar blood flow → brainstem/cerebellar ischaemia → neurological dysfunction **Clinical Features** • Vertigo and dizziness • Ataxia and gait disturbance • Diplopia • Dysarthria • Dysphagia • Nystagmus • Facial or limb sensory symptoms • Limb weakness • Drop attacks or altered consciousness in severe disease **Red flag presentation** Vertigo associated with diplopia dysarthria severe ataxia weakness or other focal neurological symptoms may indicate posterior circulation ischaemia and requires urgent medical assessment **Investigations** • MRI brain • CT brain when appropriate • CT angiography • MR angiography • Vascular Doppler studies when indicated • ECG and echocardiography when an embolic source is suspected • Glucose lipid and coagulation investigations as clinically indicated **Complications** • Transient ischaemic attacks • Posterior circulation stroke • Permanent neurological deficit • Dysphagia and aspiration • Falls • Loss of consciousness • Respiratory dysfunction in severe brainstem involvement • Death in severe stroke **Conservative / Medical Management** • Urgent stroke/TIA assessment when suspected • Antiplatelet treatment when indicated • Statin therapy when indicated • Control of blood pressure diabetes and other vascular risk factors • Smoking cessation and lifestyle modification **Physiotherapy Management** • Only after medical stabilization when appropriate • Balance and coordination training • Gait training •

Vestibular rehabilitation for suitable patients • Strengthening and functional retraining • Fall prevention • Multidisciplinary management of swallowing problems **Surgical / Endovascular Management** • Selected patients may require endovascular treatment • Angioplasty/stenting in appropriate vascular lesions • Mechanical thrombectomy for eligible acute ischaemic stroke • Surgical treatment for selected structural vascular lesions **Conclusion** Vertebrobasilar syndrome is potentially serious Recognition of posterior circulation warning signs and urgent medical referral are more important than routine musculoskeletal treatment --- **4 SCALENE SYNDROME Definition** Scalene syndrome is a form of thoracic outlet compression in which the brachial plexus and/or subclavian vessels are compressed within the interscalene triangle formed by the anterior and middle scalene muscles and the first rib **Causes** • Scalene muscle hypertrophy or spasm • Fibrous bands • Cervical rib • First rib abnormality • Poor posture • Repetitive overhead activity • Whiplash or other trauma • Heavy lifting **Pathophysiology** Scalene tightness or structural abnormality → reduced interscalene space → neurovascular compression → pain paraesthesia weakness and/or vascular symptoms **Clinical Features** • Neck and shoulder pain • Radiating arm pain • Paraesthesia and numbness • Weak grip or hand weakness • Coldness pallor or swelling when vessels are involved • Tenderness and tightness of scalene muscles • Restricted cervical movement **Investigations** • Clinical examination • Cervical/chest X ray • Doppler vascular studies when indicated • MRI/CT • EMG/NCV for neurological involvement • Vascular imaging in suspected vascular TOS **Complications** • Chronic pain • Persistent paraesthesia • Weakness and wasting • Vascular compromise • Thrombosis in vascular cases • Functional limitation **Conservative Management** • Analgesics/NSAIDs when appropriate • Activity modification • Scalene stretching • Postural correction • Deep neck flexor strengthening • Scapular stabilization • Thoracic mobility • Pectoral stretching • Breathing exercises • Neural mobilization when appropriate **Surgical Management** • Indications persistent severe symptoms progressive neurological deficit significant vascular compromise or failure of appropriate conservative treatment • Scalenectomy/scalenotomy • First rib resection • Fibrous band release • Cervical rib excision when present **Conclusion** Scalene syndrome should be differentiated from other causes of cervicobrachial pain and thoracic outlet symptoms Treatment is initially conservative unless significant neurological or vascular compromise is present --- **5 COSTOCLAVICULAR SYNDROME Definition** Costoclavicular syndrome is a form of thoracic outlet syndrome caused by compression of neurovascular structures between the clavicle and first rib when the costoclavicular space becomes narrowed **Causes** • Drooping shoulders and poor posture • Clavicular deformity or fracture malunion • First rib abnormality • Surrounding muscle tightness • Carrying heavy loads • Trauma • Repetitive shoulder depression **Pathophysiology** Shoulder depression or structural abnormality → clavicle approaches first rib → costoclavicular space narrows → neurovascular compression → neurological and/or vascular symptoms **Clinical Features** • Pain in neck shoulder or arm • Paraesthesia and numbness • Weakness and hand fatigue • Arm heaviness • Swelling or cyanosis in venous involvement • Coldness pallor or reduced circulation in arterial involvement • Symptoms may be reproduced by costoclavicular positioning **Investigations** • X ray • Doppler ultrasound • CT/MR angiography when vascular disease is suspected • MRI • EMG/NCV for selected neurological cases **Complications** • Chronic nerve compression • Weakness • Venous thrombosis • Arterial compromise • Ischaemia • Functional impairment **Conservative Management** • Analgesics and activity modification • Postural correction • Scapular stabilization • Shoulder girdle strengthening • Pectoral stretching • Thoracic extension and mobility exercises • Breathing exercises • Ergonomic correction • Avoidance of heavy loads **Surgical Management** • First rib resection in selected severe cases • Decompression of abnormal structures • Clavicular decompression in selected structural cases • Vascular reconstruction when required **Conclusion** The objective is to increase the costoclavicular space reduce neurovascular compression and restore normal shoulder girdle mechanics --- **6 LEVATOR SCAPULAE SYNDROME Definition** Levator scapulae syndrome is a painful musculoskeletal condition characterized by pain tenderness and increased tension or dysfunction of the levator scapulae muscle commonly associated with poor posture repetitive activity and cervical/scapular dysfunction **Causes** • Forward head posture • Rounded shoulders • Prolonged sitting • Poor workstation ergonomics • Levator scapulae overuse • Muscle spasm or trigger points • Whiplash or sudden neck movement • Cervical spondylosis • Scapular dyskinesis • Repetitive overhead activity **Pathophysiology** Prolonged postural loading or overuse → increased levator scapulae tension → local tenderness/trigger points → pain and restricted cervical/scapular movement **Clinical Features** • Pain at the superomedial angle of the scapula • Neck and upper shoulder pain • Pain toward the medial scapular border • Local tenderness and trigger points • Restricted cervical ROM • Pain with prolonged sitting • Symptoms aggravated by cervical rotation or side flexion **Investigations** • Usually a clinical diagnosis • Postural and scapular movement assessment • Cervical X ray when indicated • MRI if significant cervical pathology is suspected **Complications** • Chronic neck pain • Recurrent muscle spasm • Headache • Reduced cervical ROM • Scapular dysfunction • Reduced work tolerance • Sleep disturbance **Conservative / Physiotherapy Management** • Heat or TENS for symptom relief when appropriate • Soft tissue and manual therapy • Levator scapulae and upper trapezius stretching • Pectoral stretching • Deep neck flexor strengthening • Scapular retractors lower trapezius and serratus anterior strengthening • Postural correction • Ergonomic modification • Trigger point

techniques • Home exercise programme **Surgical Management** Surgery is rarely required If symptoms are due to an underlying structural cervical or shoulder disorder treatment is directed toward the underlying pathology **Conclusion** Most patients respond well to correction of posture muscle imbalance mobility restrictions and ergonomic factors --- **7**

PIRIFORMIS SYNDROME Definition Piriformis syndrome is a condition in which the sciatic nerve is irritated or compressed by the piriformis muscle in the deep gluteal region producing buttock pain with or without radiating sciatic type symptoms **Causes** • Anatomical variation between piriformis and sciatic nerve • Piriformis hypertrophy or spasm • Direct buttock trauma • Prolonged sitting • Running or repetitive hip rotation • Overuse • Poor biomechanics • Sacroiliac or hip dysfunction **Pathophysiology** Piriformis overuse/spasm or anatomical variation → sciatic nerve irritation/compression → local inflammation → buttock pain with or without radiating symptoms **Clinical Features** • Deep buttock pain • Pain radiating into posterior thigh • Sciatic type symptoms • Paraesthesia or numbness • Pain aggravated by prolonged sitting • Tenderness over piriformis • Pain with resisted hip external rotation • Pain with piriformis stretching • Possible reduced hip ROM **Special Tests** • FAIR test • Freiberg sign • Pace sign • Beatty manoeuvre **Differential Diagnosis** • Lumbar radiculopathy • Lumbar disc prolapse • Sacroiliac joint dysfunction • Hip pathology • Hamstring pathology • Spinal stenosis **Investigations** • Primarily a clinical diagnosis • MRI lumbar spine or pelvis to exclude other pathology • EMG/NCV in selected cases • Ultrasound in selected cases • Diagnostic local injection in selected persistent cases **Complications** • Chronic buttock pain • Persistent sciatic symptoms • Reduced walking and sitting tolerance • Sleep disturbance • Reduced athletic performance • Functional disability **Conservative / Physiotherapy Management** • Activity modification and avoidance of prolonged sitting • Heat/cold or TENS when appropriate • Piriformis hip rotator and gluteal stretching • Gluteus maximus and medius strengthening • Core strengthening • Neural mobilization when appropriate • Gait/running biomechanics correction • Gradual return to sport • Analgesics/NSAIDs when appropriate • Selected image guided injection for persistent symptoms **Surgical Management** • Rarely required • Considered after appropriate conservative treatment in severe persistent cases with convincing evidence of sciatic nerve compression • Piriformis release • Sciatic nerve decompression • Correction of associated structural abnormalities **Conclusion** A careful differential diagnosis is essential because lumbar radiculopathy and other hip/spinal disorders can mimic piriformis syndrome/// **CO4 ///** -----

-- **1 CEREBRAL PALSY Definition** Cerebral palsy is a group of permanent disorders of movement and posture causing activity limitation attributed to non progressive disturbances occurring in the developing fetal or infant brain It may be accompanied by disturbances of sensation perception cognition communication behaviour and epilepsy **Causes / risk factors** • Prenatal congenital brain malformations intrauterine infections genetic abnormalities maternal diabetes or hypertension and placental abnormalities • Perinatal prematurity low birth weight hypoxic ischaemic injury intracranial haemorrhage and neonatal seizures • Postnatal meningitis encephalitis head injury cerebrovascular injury severe hypoxia kernicterus and hypoglycaemia **Classification** • Spastic CP hemiplegia diplegia and quadriplegia • Dyskinetic CP dystonia and choreoathetosis • Ataxic CP poor balance incoordination and tremor • Mixed CP combination of motor patterns **Clinical features** • Delayed motor milestones • Abnormal muscle tone spasticity dystonia or hypotonia • Weakness and poor selective motor control • Abnormal reflexes and persistence of primitive reflexes • Abnormal gait and poor balance • Contractures and deformities • Hip displacement scoliosis and foot deformities • Seizures • Visual/hearing impairment • Speech communication feeding or swallowing difficulties • Cognitive or behavioural difficulties in some individuals **Complications** • Contractures • Joint deformities • Hip subluxation/dislocation • Scoliosis • Osteopenia • Pain • Reduced mobility • Epilepsy • Feeding and respiratory problems • Reduced independence and participation **Investigations** • Neurological and developmental assessment • MRI brain • EEG when seizures are suspected • Hip radiographs when indicated • Spine radiographs when indicated • Vision and hearing assessment • Functional assessment **Medical management** • Individualized multidisciplinary management • Spasticity oral antispastic medications in selected patients • Botulinum toxin for focal spasticity • Intrathecal baclofen in selected severe cases • Antiepileptic medication when required • Pain feeding constipation bladder nutrition and respiratory management **Surgical management** • Tendon lengthening and tendon transfer • Osteotomy and correction of hip displacement • Foot reconstruction • Spinal surgery for severe scoliosis • Selective dorsal rhizotomy in carefully selected children with severe lower limb spasticity • Intrathecal baclofen pump in selected severe generalized spasticity **Physiotherapy management** • Promote motor development and functional movement • Positioning and seating for alignment • Passive/active ROM and stretching • Strengthening and weight bearing activities • Postural control and balance training • Gait training with appropriate aids • Orthoses when indicated • Transfer ADL and mobility training • Respiratory physiotherapy when required • Home exercise and caregiver education **Multidisciplinary approach** • Paediatrician/neurologist • Orthopaedic surgeon • Physiotherapist • Occupational therapist • Speech and language therapist • Orthotist • Psychologist • Special educator • Dietitian • Social worker and rehabilitation physician **Conclusion** Cerebral palsy is a lifelong condition requiring early individualized and multidisciplinary rehabilitation Management aims to maximize mobility independence participation and quality of life while preventing secondary complications --- **2 POLIOMYELITIS Definition** Poliomyelitis is an acute viral

infectious disease caused by poliovirus primarily affecting motor neurons of the anterior horn of the spinal cord and sometimes the brainstem resulting in acute flaccid paralysis

Causative organism and transmission • Poliovirus is an enterovirus of the Picornaviridae family • Transmission is mainly feco oral through contaminated food and water • Oral transmission may also occur

Pathogenesis • Virus enters through gastrointestinal or respiratory tract • Replicates in pharynx and intestine • May enter bloodstream • In some individuals reaches the CNS • Motor neurons of anterior horn may be destroyed • This produces lower motor neuron weakness and flaccid paralysis

Clinical types • Inapparent infection • Abortive poliomyelitis fever malaise headache sore throat nausea/vomiting • Non paralytic poliomyelitis meningitic symptoms such as headache neck stiffness and muscle pain • Paralytic poliomyelitis acute flaccid often asymmetrical paralysis with reduced/absent reflexes • Bulbar poliomyelitis dysphagia dysarthria respiratory difficulty and cranial nerve dysfunction

Clinical features • Fever fatigue and headache • Nausea/vomiting • Muscle pain • Acute flaccid weakness • Asymmetrical paralysis • Reduced muscle tone • Reduced/absent deep tendon reflexes • Muscle wasting • Respiratory and swallowing weakness in severe disease • Sensation is generally preserved

Complications • Respiratory failure • Aspiration • Permanent paralysis • Contractures • Joint deformities • Scoliosis • Limb length discrepancy • Osteoporosis • Gait abnormalities • Post polio syndrome

Investigations • Clinical neurological examination • Stool testing for poliovirus • Throat specimens when appropriate • CSF examination • Electromyography • Nerve conduction studies • Respiratory function assessment when required

Medical management • There is no specific treatment that reverses established paralytic poliomyelitis • Supportive care hydration nutrition and pain/fever management • Respiratory support and assisted ventilation when required • Airway clearance • Swallowing management • Prevention and treatment of complications

Surgical management • Surgery does not treat the acute viral infection • For residual deformity tendon transfer tendon lengthening osteotomy contracture correction foot reconstruction and spinal surgery for severe scoliosis

Physiotherapy management • Acute positioning gentle passive ROM breathing exercises pressure relieving measures and appropriate splinting • Avoid excessive exercise of acutely denervated muscles • Recovery active assisted and gradual active exercise muscle re education and strengthening of useful innervated muscles • Residual gait training orthoses functional independence and energy conservation • Post polio syndrome pacing energy conservation appropriate low/moderate exercise and avoidance of overuse

Prevention • Vaccination with IPV and/or OPV according to the applicable immunization programme • Safe water and sanitation • Hand hygiene • Surveillance and outbreak control

Conclusion Poliomyelitis can produce permanent motor disability through anterior horn motor neuron injury Supportive medical care prevention of secondary complications physiotherapy orthotic management and reconstructive surgery when necessary are central to rehabilitation

--- **3 SPINAL DYSRAPHISM Definition** Spinal dysraphism is a group of congenital disorders caused by abnormal development and incomplete closure of neural tube and/or vertebral structures producing defects involving the spinal cord meninges vertebrae and overlying tissues

Classification • Open dysraphism neural tissue is exposed or incompletely covered • Myelomeningocele spinal cord and meninges protrude through a vertebral defect • Myelocele neural tissue is directly exposed • Closed dysraphism skin covered defects such as spina bifida occulta meningocele lipomyelomeningocele tethered cord and dermal sinus

Causes / risk factors • Multifactorial origin • Folate deficiency • Maternal nutritional deficiency • Certain maternal medications • Maternal diabetes • Genetic susceptibility • Previous pregnancy affected by neural tube defect • Environmental factors

Clinical features • Lower limb weakness or paralysis • Flaccid or spastic paralysis depending on lesion • Reduced sensation • Altered reflexes • Bladder and bowel dysfunction • Hip dislocation • Contractures • Clubfoot and other foot deformities • Knee deformities • Scoliosis/kyphosis • Hydrocephalus • Chiari II malformation especially with myelomeningocele • Tethered cord • Pressure sores

Complications • Hydrocephalus • Neurological deterioration • Tethered cord syndrome • Recurrent urinary tract infection • Renal damage • Bowel dysfunction • Contractures and deformities • Pressure sores • Scoliosis • Reduced mobility • Psychosocial difficulties

Investigations • Prenatal ultrasound and maternal screening • Fetal MRI in selected cases • Postnatal MRI spine • Ultrasound/CT in selected cases • Neurological examination • Urodynamic studies • Renal function assessment • Orthopaedic assessment

Medical management • Hydrocephalus management • Bladder and bowel management • Prevention of infection • Skin care • Nutritional support • Pain management • Management of associated neurological problems

Surgical management • Early closure of myelomeningocele to protect neural tissue • Hydrocephalus ventriculoperitoneal shunt or endoscopic third ventriculostomy in selected cases • Tethered cord release when clinically indicated • Orthopaedic procedures tendon procedures osteotomy foot correction hip procedures and spinal deformity correction

Physiotherapy management • Protect surgical wound and prevent pressure sores • Proper positioning and gentle ROM • Facilitate motor development • Improve head and trunk control • Strengthen available muscles • Encourage age appropriate mobility • Stretching splinting and orthotic management • Sitting and standing balance • Gait training when possible • Wheelchair and transfer skills • ADL and community mobility training

Bladder and bowel rehabilitation • Timed voiding • Clean intermittent catheterization when prescribed • Bowel programme • Skin care • Patient and family education

Multidisciplinary approach • Neurosurgeon • Paediatrician • Urologist • Orthopaedic surgeon • Physiotherapist

LEPROSY Definition Leprosy Hansen disease is a chronic infectious disease caused mainly by *Mycobacterium leprae* primarily affecting the skin peripheral nerves mucosa of the upper respiratory tract and eyes **Causative organism and transmission** • *Mycobacterium leprae* is an acid fast bacillus • Primarily affects Schwann cells and macrophages • Transmission is associated with prolonged close contact with untreated infectious disease particularly respiratory droplets • It is not spread by casual brief contact **Classification** • WHO operational classification paucibacillary PB and multibacillary MB • Ridley Jopling spectrum indeterminate tuberculoid borderline tuberculoid mid borderline borderline lepromatous and lepromatous **Clinical features** • Hypopigmented or reddish skin patches • Definite sensory loss over lesions • Dry skin and loss of sweating • Nodules in some forms • Loss of hair over lesions • Thickened peripheral nerves with sensory/motor deficit • Paraesthesia and numbness • Muscle weakness and wasting • Foot drop • Claw hand • Lagophthalmos • Recurrent ulcers and burns **Common nerves affected** • Ulnar nerve • Median nerve • Radial nerve • Common peroneal nerve • Posterior tibial nerve • Facial nerve **Complications** • Peripheral neuropathy • Sensory loss and paralysis • Contractures • Claw hand • Foot drop • Muscle wasting • Recurrent ulcers • Burns and secondary infection • Corneal anaesthesia and ulceration • Visual impairment • Stigma and social/occupational difficulties **Investigations** • Complete skin examination • Sensory testing • Peripheral nerve examination • Motor examination • Slit skin smear when indicated • Skin biopsy • Eye examination • Functional assessment **Medical management** • Multidrug therapy MDT is the cornerstone • Common antileprosy drugs rifampicin dapsone and clofazimine • Regimen and duration depend on classification and current national/WHO guidance • Early treatment helps prevent nerve damage and disability **Management of reactions** • Type 1 reaction may cause inflammation and acute nerve impairment • Type 2 reaction may cause fever painful nodules and systemic inflammation • Corticosteroids and other specialist directed treatment may be required depending on the reaction **Surgical management** • Selected nerve decompression • Tendon transfer for claw hand • Tendon transfer for foot drop • Contracture correction • Reconstructive procedures • Ulcer management • Eyelid procedures for lagophthalmos **Physiotherapy management** • Assess sensation muscle power ROM deformity skin integrity balance and gait • Daily skin inspection and protective footwear • Avoid extreme heat and barefoot walking • ROM stretching and strengthening of functioning muscles • Balance and functional training • Claw hand splinting ROM strengthening and post tendon transfer rehabilitation • Foot drop AFO gait training and balance training • Ulcer prevention pressure relief off loading appropriate footwear and education **Multidisciplinary approach** • Dermatologist/physician • Neurologist • Physiotherapist • Occupational therapist • Orthopaedic/plastic surgeon • Ophthalmologist • Foot care services where available • Psychologist and social worker **Prevention** • Early diagnosis • Prompt MDT treatment • Contact examination/surveillance • Appropriate preventive strategies for eligible contacts • Health education • Early recognition of nerve involvement • Disability prevention and self care education **Conclusion** Leprosy is a chronic infectious disease with important neurological and disability related consequences Early diagnosis and MDT prevent progression while physiotherapy orthoses reconstructive surgery and self care education help prevent disability and maintain function ///